ITA0329-MOBILE COMPUTING

ASSIGNMENT-1

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ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual

Duration: 1 Week

Total Marks: 50

Submission Date: _____

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration: 1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.
- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least three meaningfully different strategies (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an iterative procedure (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a recursive/hierarchical procedure (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the access channels are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

TASK 2 – Identify the Governing Design Parameter 2 Marks

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km².
- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine planning cost (survey, number of sites) with operating capacity to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

TASK 5 – Scaling and Sensitivity Analysis 3 Marks

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round $k - 1$?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

TASK 7 – Solve the Recurrence and Interpret the Result 4 Marks

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4 (D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

TASK 8 – Implement and Test the Planning Strategies 4 Marks

Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

TASK 9 – Record and Present Experimental Performance 3 Marks

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

TASK 10 – Compare Theory with Simulation 4 Marks

Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 11 – Recommend and Defend the Final Design 4 Marks

Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

C. Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

D. Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

MARK DISTRIBUTION

Task	Marks
Task 1 – Design Alternative Frequency-Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6
Task 4 – Analyse Best, Average and Worst-Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3
Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4
Task 8 – Implement and Test the Planning Strategies	4
Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
Subtotal	42
Task 12 – Day-4 Design Challenge	8
TOTAL	50

ANSWERS

The answers below follow the order of the tasks. Every number quoted comes either from a derivation shown in the text or from the simulator described in Task 8 (Python/NumPy, fixed seeds, results reproducible). Strategy labels used throughout: **S1** fixed cluster reuse, **S2** sectorised reuse, **S3** hierarchical cell splitting (macro–micro), **S4** dynamic/borrowed channel assignment, and **S2+S3** the hybrid of S2 and S3.

Common planning data and adopted parameters

Demand. Total busy-hour traffic is $A = a \cdot U$ with $a = 0.025$ Erlang per subscriber. With a mean call holding time of 120 s, one Erlang corresponds to 30 call attempts per busy hour. The township area is 12 km², so the average demand density is A/12.

Table 1: Forecast demand over the planning horizon

Subscribers U	Offered traffic (Erlang)	Density (E/km²)	Call attempts / busy hour
1,000	25	2.1	750
10,000	250	20.8	7,500
20,000	500	41.7	15,000
100,000	2,500	208.3	75,000
200,000	5,000	416.7	150,000

SIR threshold: 15 dB. A digital air interface with channel coding, equalisation and frequency hopping can work at about 12 dB, whereas an analogue-style system needs about 18 dB for acceptable quality. The township is a mostly pedestrian, low-speed environment but has dense buildings, so shadowing is severe and imperfect power control must be allowed for. I therefore adopt $\gamma = 15$ dB, which is 3 dB above the digital lower bound (a margin for shadowing and fading) and 3 dB below the analogue figure (so that spectrum is not wasted on a requirement the technology does not have). The choice matters: at $n = 3.5$ the minimum cluster size is $N = 7$ for both 12 dB and 15 dB but jumps to $N = 12$ at 18 dB, which would cut the capacity per cell by almost half (38.4 to 20.2 Erlang).

Path-loss exponents. $n = 2.5$ for the open campus (quadrangles, sports grounds, open land, line-of-sight dominated); $n = 3.0$ for the residential colonies and the arterial road (low-rise suburban clutter); $n = 4.0$ for the dense academic blocks and hostels (obstruction, indoor penetration, street canyons). Weighting by where subscribers actually live (15 % open, 35 % suburban, 50 % dense) gives $0.15 \times 2.5 + 0.35 \times 3.0 + 0.50 \times 4.0 = 3.4$, so the **design value is n = 3.5** (slightly conservative), and every result is also reported for 2.5, 3 and 4. Log-normal shadowing of $\sigma = 8$ dB (typical urban value) is used wherever shadowing is simulated.

Channels. The 336 duplex voice channels are treated as 42 carriers × 8 TDMA time slots (a GSM-like organisation: $336 = 42 \times 8$), so a cell with 48 channels owns 6 carriers. The control-channel overhead has already been removed from the 336, so it is the traffic pool that is shared between cells.

Illustrative cost model (the question gives no prices, so the following assumptions are stated explicitly). Propagation survey (drive test) Rs 0.8 lakh per km²; traffic estimation study Rs 3 lakh (one-off); radio planning and site-acquisition work Rs 0.6 lakh per site; site capital cost $C_s(R) = 30 \cdot \sqrt{R}$ lakh per omni site (R in km, with a floor at R = 60 m), multiplied by 1.25 for a three-sector site; 1 crore = 100 lakh. A cell of smaller radius is cheaper because it needs a shorter mast and less transmit power, hence the √R dependence.

Answer to Task 1 – Design Alternative Frequency-Planning Strategies (5 Marks)

A fixed block of S = 336 channels can be spread over the township in fundamentally different ways. Four are proposed here because each uses a different physical mechanism to reconcile capacity with interference: **S1** separates co-channel cells by distance (fixed reuse); **S2** separates them by direction (antenna sectoring); **S3** shrinks the cells so the same channels are reused more often in space (hierarchical splitting); **S4** removes the fixed assignment and pools the channels (dynamic borrowing). Changing the cluster size inside S1 does not create a new strategy.

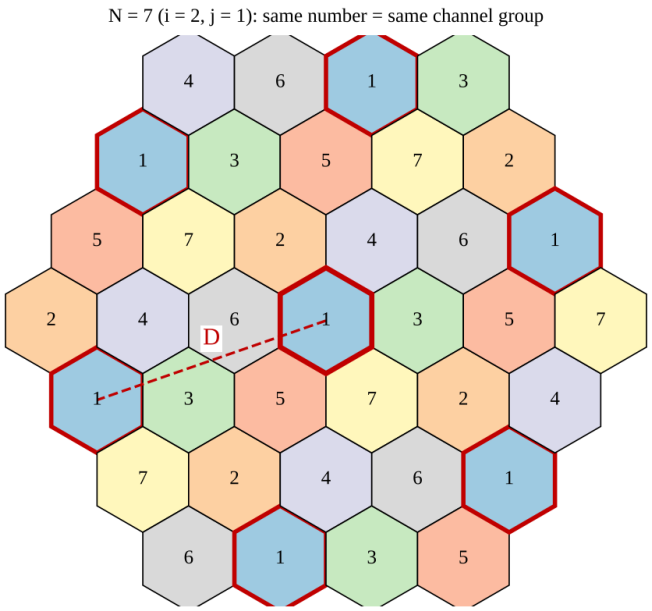


Figure 1: Hexagonal cell, cluster of N = 7 and reuse distance D. Cells with the same number share one channel group; the six red cells are the first-tier co-channel interferers.

1.1 Strategy S1 – Fixed cluster reuse (iterative procedure)

Basic idea. Cells are grouped into clusters of N cells, $N = i^2 + ij + j^2$. The S channels are divided into N disjoint groups of S/N channels, one group per cell of the cluster, and the same pattern is repeated across the township. Two cells using the same group are exactly $D = R\sqrt{(3N)}$ apart. The smallest cluster whose SIR meets the threshold is selected, because capacity falls as 1/N.

Assumptions. Regular hexagonal cells of equal radius R; uniform traffic; omnidirectional antennas; six equidistant first-tier interferers; fixed (static) channel assignment; blocked calls cleared and Poisson arrivals so that Erlang-B applies.

Access channels. Each of the S/N channels is a traffic channel in an FDMA/TDMA scheme (carrier × slot). One carrier of every cluster-group also carries the broadcast and signalling (control) channels, so control channels are reused on the same pattern and are protected by the same SIR. A CDMA system would instead use N = 1 and be limited by total interference (soft capacity); in OFDMA the same idea appears as fractional frequency reuse (N = 3 at the cell edge, N = 1 near the base station).

Pseudocode S1

```
ALGORITHM S1(S, n, gamma_dB, A_total, Area, GoS)
 gamma <- 10^(gamma_dB/10)
 for each valid N in increasing order (N = i^2 + ij + j^2):   # iterative search
    Q   <- sqrt(3*N)                # reuse ratio D/R
    SIR <- Q^n / 6                  # six first-tier interferers
    if SIR >= gamma: break
 c     <- floor(S / N)              # channels per cell
 A_cell <- ErlangB_inverse(c, GoS)     # Erlang carried at 2% blocking
 M     <- ceil(A_total / A_cell)      # cells needed
 R     <- sqrt(Area / (2.598 * M))    # radius from area balance
 for each cell (q, r) of the hex grid:
    group(q, r) <- (a*q + b*r) mod N  # co-channel cells D apart
    give the cell its c channels of that group
 return N, c, M, R
```

Design principle: spatial (frequency) reuse combined with trunking (a pool of c channels shared by all subscribers of a cell). **Why investigate it:** it is the baseline of every cellular system; it exposes the central trade-off (small N gives capacity, large N gives SIR) in its purest form, and it is the reference against which the other strategies are judged.

1.2 Strategy S2 – Sectorised reuse (iterative over sector count)

Basic idea. Each omni cell is replaced by s = 3 (120°) or s = 6 (60°) sectors fed by directional antennas, and each sector is treated as a cell with its own channel set of S/(N·s) channels. A mobile is illuminated by only the co-channel sectors that point at it, so the number of first-tier interferers falls from 6 to about 2 (120°) or 1 (60°). The saved interference is spent on a smaller N, which raises the capacity per cell.

Assumptions. Ideal sector beams with a finite front-to-back ratio (20 dB is used in the simulator); identical orientation of sectors in all cells; handoffs between sectors of the same site are cheap; the channel pool is split among sectors, so trunking efficiency falls (this is the price of sectoring).

Access channels. As in S1 (FDMA/TDMA), but the control channels are repeated in every sector, so each sector needs its own broadcast carrier. In CDMA, sectoring raises capacity directly by cutting interference by roughly a factor s.

Pseudocode S2

```
ALGORITHM S2(S, n, gamma_dB, A_total, Area, GoS)
 best <- none
 for s in {3, 6}:                    # iterate over sector count
    i0 <- 2 if s = 3 else 1          # interferers seen by a sector
    N  <- smallest valid cluster size with Q^n / i0 >= gamma
    c  <- floor(S / (N * s))         # channels per sector
    A_cell <- s * ErlangB_inverse(c, GoS)
    M  <- ceil(A_total / A_cell)
    cost <- planning_cost(M) + M * sector_site_cost(s, R(M))
    keep (s, N, c, M, cost) with the lowest cost
 return best
```

Design principle: spatial filtering (directional antennas) plus reuse; trunking efficiency is deliberately sacrificed. **Why investigate it:** sectoring needs no new sites, only new antennas, so it is the cheapest way to raise capacity of an existing grid, and it directly attacks the dominant impairment (the number of interferers).

1.3 Strategy S3 – Hierarchical macro–micro layering by cell splitting (recursive procedure)

Basic idea. Start with a coarse layer of large macro cells (R_0 = 1 km) that already gives full coverage and uses a fixed N = 7 channel plan. Whenever a cell is offered more traffic than its c channels can carry at 2 % blocking, it is split into four cells of radius R/2 (three new base stations are added), transmit power is cut to $P' = P/2^n$, and the process is repeated on the new cells until no cell is overloaded. Because R/N is the same, D/R and therefore the SIR do not change.

Assumptions. The channel plan N = 7 is fixed from day one so that no re-planning is needed; daughter cells reuse the parent group; only overloaded cells are split (selective splitting) so the network is a mixture of cell sizes, with the channels of a split cell divided between the larger and smaller layers when both coexist; small cells have low mobility demand.

Access channels. FDMA/TDMA as in S1. The control carrier is reused in the daughter cells, so there is no extra control overhead per round, but the signalling load grows with the handoff rate (Task 7).

Pseudocode S3

```
PROCEDURE SPLIT(cells, k)                # recursive
  over <- { c in cells : offered(c) > ErlangB_inverse(S/7, 2%) }
  if over is empty or R(c) < R_min: return cells, k     # base case
  for c in over:
     replace c by 4 cells of radius R(c)/2
     P(new) <- P(c) / 2^n                # keep the cell-edge power
     assign the parent's channel group to all four cells
  return SPLIT(cells', k + 1)

main: cells <- hex grid with R0 = 1 km; SPLIT(cells, 0)
```

Design principle: spatial reuse obtained by scaling the cell size (divide and conquer on the map); the trunking efficiency of each cell is kept at the N = 7 value. **Why investigate it:** it is the only strategy that matches the capacity to where the demand is, which matters because a university township has strong hot spots (academic blocks, hostels), and it gives a clean growth path with no re-planning.

1.4 Strategy S4 – Dynamic / borrowed channel assignment

Basic idea. Channels are not owned by cells. A call that finds its own channels busy borrows a free channel from a neighbouring cell, or takes any channel of the whole pool, provided the minimum reuse distance $D_{min} = R(6\gamma)^{1/n}$ from every cell currently using that channel is respected. The reuse distance is thus adapted to the actual propagation and traffic instead of being fixed by an integer N.

Assumptions. A central controller (MSC) knows the occupancy and interference of all channels; measurements of the local path-loss exponent are available; borrowed channels may lock co-channel neighbours; a reserve of channels is kept for the owner so that borrowing does not starve it.

Access channels. Any FDMA/TDMA slot of the common pool can serve any cell; the control channels remain fixed so that handsets can always find a cell.

Pseudocode S4

```
ON call arrival in cell x:
  if x has a free own channel: assign it
  else:
    for each neighbour y of x (richest first):
       if y has a free channel ch and busy(y) <= (1 - reserve)*c:
          if every cell using ch is at least D_min from x:
             borrow ch; lock co-channel cells within D_min; return OK
    block the call
ON call end: release ch and unlock the cells it had locked
```

Design principle: trunking (pooling of channels across cells) with dynamic spatial reuse. **Why investigate it:** it removes the integer rounding of N and the idle channels of a lightly loaded cell, and it is the natural tool for temporary, unpredictable surges, so it is needed for the Day-4 challenge.

1.5 Preprocessing required and who bears the cost

Table 2: Preprocessing and cost bearer

Preprocessing step	Needed by	Bearer
Traffic map (subscriber density by block, hot spots, arterial-road flow)	S1 (average only), S2, S3 (essential for selective splitting), S4 (statistics)	Operator, with occupancy data supplied by the university
Propagation survey (drive test) to fix n and σ	All; S4 also needs live measurements	Operator (or contractor)
Hex-grid layout and site acquisition	All; S3 repeats it for each round	Operator; university grants rooftops/land
Frequency plan (channel groups, cluster colouring)	S1, S2, S3 once; S4 not needed	Operator; regulator allocates the 336 channels
Controller software and interference measurement	S4 only	Operator (equipment vendor)

All of this effort is counted in the cost model of Task 3: Rs 9.6 lakh for the 12 km² survey, Rs 3 lakh for the traffic study and Rs 0.6 lakh per site for planning. The numbers of sites that drive the last item are produced by the strategies themselves, which is why strategy choice and planning cost cannot be separated.

Summary of the answer. Four distinct strategies, S1 (iterative), S3 (recursive), S2 (spatial filtering) and S4 (dynamic pooling), have been defined, each with an access-channel description, pseudocode, a design principle and a reason for investigation. S1, S2 and S3 are analysed and simulated in full; S4 is analysed through the Erlang-B pooling argument and a call-level simulation.

Answer to Task 2 – Identify the Governing Design Parameter (2 Marks)

Table 3: Governing design parameter of each strategy

Strategy	Governing parameter	Where it appears	Why it governs
S1 Fixed cluster	Cluster size N (equivalently $Q = D/R = \sqrt{(3N)}$)	Channels per cell S/N; $SIR = Q^n/6$	One number sets both capacity ($\propto 1/N$) and interference ($SIR \propto N^{n/2}$)

Strategy	Governing parameter	Where it appears	Why it governs
S2 Sectorised	Number of interferers i_o (6, 2 or 1) together with N	SIR = Q^n/i_o; channels per sector S/(sN)	Sectoring changes i_o without changing N; capacity gain comes from the smaller N that i_o allows
S3 Splitting	Cell radius R (round k: $R_k = R_o/2^k$)	Cells $\propto 1/R^2$, power $\propto R^n$, handoffs $\propto 1/R$	SIR is unchanged by splitting, so R alone sets capacity, power and mobility load
S4 Dynamic	Minimum reuse distance $D_{min} = R(6\gamma)^{1/n}$ and pool size c	Channel admission test; Erlang-B on the pooled channels	It is the integer N replaced by a continuous, locally adapted quantity

Why the analysis is built on it. For S1 the ratio Q is the only geometric quantity that appears in the SIR, and $N = Q^2/3$ is the only quantity that appears in the channels-per-cell expression, so every efficiency question (capacity per km², cells needed, cost per Erlang) can be written as a function of N alone. For S2 the number of interferers i_o plays the same role, since sectoring leaves Q untouched and only changes the divisor in the SIR. For S3 the radius is the natural variable because splitting by definition changes R and nothing else in the SIR expression. For S4 the effective cluster size $N_{eff} = (6\gamma)^{2/n}/3$ is a real number, 6.67 at n = 3.5 instead of the integer 7 and 4.59 at n = 4 instead of 7.

How the parameter must change as subscribers grow from 1,000 to 100,000+. N and i_o are fixed by the propagation environment and the threshold, not by demand: with γ = 15 dB and n = 3.5, S1 must keep N = 7 and S2 must keep N = 4 (120°) whatever the subscriber count. Growth in demand therefore has to be absorbed by the second parameter, the radius R. From cells $M = A_{tot}/(2.598R^2)$ and $M = aU/A_{cell}$, $R_{req} = \sqrt{(A_{tot}/(2.598 \cdot M))}$ falls as $U^{-1/2}$. For S1 with N = 7 (38.4 Erlang per cell) the uniform-demand radius is 1.24 km (coverage limited, 3 cells) at 1,000 subscribers, 0.81 km at 10,000, 0.57 km at 20,000 and 0.26 km at 100,000. For S3 the radius moves in halving steps, R = 1.0, 0.5, 0.5 and 0.25 km for the four forecasts (k = 0, 1, 1, 2 rounds, Task 7). For S4 the governing quantity D_{min} does not move with demand at all; the gain is that borrowing lets each cell use the channels that others leave idle.

Answer to Task 3 – Derive the Capacity and Interference Functions Mathematically (6 Marks)

3.1 Hexagonal geometry, cluster size and reuse distance

Cell area. A regular hexagon of circumradius R is made of six equilateral triangles of side R. Each triangle has height $R \cdot \sin 60° = (\sqrt{3}/2)R$ and area $\frac{1}{2} \cdot R \cdot (\sqrt{3}/2)R = (\sqrt{3}/4)R^2$. Hence

$$A_c = 6 \cdot (\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 \approx 2.598\ R^2$$

The distance between the centres of two adjacent cells is twice the apothem, $2\cdot(\sqrt{3}/2)R = \sqrt{3}\cdot R$.

Cluster size. To move from a cell to the nearest cell of the same group, move i cell-steps along one axis, turn 60° anticlockwise and move j steps along the second axis. The two legs have lengths $i\sqrt{3}R$ and $j\sqrt{3}R$ and enclose an angle of 120° (the interior angle of the triangle they form with the line D). By the cosine rule

$$D^2 = (i\sqrt{3}R)^2 + (j\sqrt{3}R)^2 - 2(i\sqrt{3}R)(j\sqrt{3}R)\cos 120° = 3R^2(i^2 + ij + j^2)$$

The cluster size is therefore $N = i^2 + ij + j^2$ (1, 3, 4, 7, 9, 12, 13, 16, 19, 21, 25, 27, …) and

$$D = R\sqrt{(3N)}, \quad Q = D/R = \sqrt{(3N)}$$

Check by area: the co-channel lattice cell is a rhombus of side D and angle 60°, of area $D^2 \sin 60° = (\sqrt{3}/2)D^2$. It must contain exactly N hexagonal cells, so $(\sqrt{3}/2)D^2 = N\cdot(3\sqrt{3}/2)R^2$, which gives $D^2 = 3NR^2$ again.

Table 4: Valid cluster sizes with reuse ratio, channels, traffic and omni SIR

N	Q = √(3N)	Channels / cell	Erlang / cell at 2 %	SIR n = 3.5 (dB)	SIR n = 4 (dB)
3	3.00	112	99.6	8.9	11.3
4	3.46	84	72.5	11.1	13.8
7	4.58	48	38.4	15.4	18.7
9	5.20	37	28.3	17.3	20.8
12	6.00	28	20.2	19.5	23.3
13	6.24	25	17.5	20.1	24.0
19	7.55	17	10.7	22.9	27.3
21	7.94	16	9.8	23.7	28.2

Number of cells to cover 12 km². Neglecting boundary effects, $M = A_{tot}/A_c = 12/(2.598R^2) = 4.62/R^2$ (R in km), rounded up. R = 1 km needs 5 cells, R = 0.5 km needs 19, R = 0.25 km needs 74 and R = 0.125 km needs 296. The simulator, which has to cover a 4 km × 3 km rectangle with whole hexagons, needs 7 cells at R = 1 km because of the edges (Task 10 explains this difference).

3.2 Channels per cell, traffic per cell and system capacity

Channels per cell. Dividing the S = 336 channels equally among the N cells of a cluster gives $c = \lfloor S/N \rfloor$. For sectored cells $c = \lfloor S/(sN) \rfloor$ per sector. Rounding loses (S mod N) channels per cluster, up to 3.9 % for N = 19 (17 channels × 19 = 323).

Traffic per cell, Erlang-B. With Poisson arrivals, exponentially distributed holding times, c servers and blocked calls cleared, the state probabilities of the birth–death chain are proportional to $A^k/k!$. Calls are blocked when all c channels are busy, so

$$B(A, c) = (A^c / c!) / \Sigma_{k=0}^{c} (A^k / k!)$$

For numerical use the recursion $B(A,0) = 1$, $B(A,k) = A \cdot B(A,k-1)/(k + A \cdot B(A,k-1))$ is exact and stable. The traffic a cell can carry is the inverse $A_{cell} = B^{-1}(0.02, c)$, found by bisection (this is what the simulator does). The carried traffic is $A(1 - B)$, and the trunking efficiency is $\eta = A_{cell}/c$.

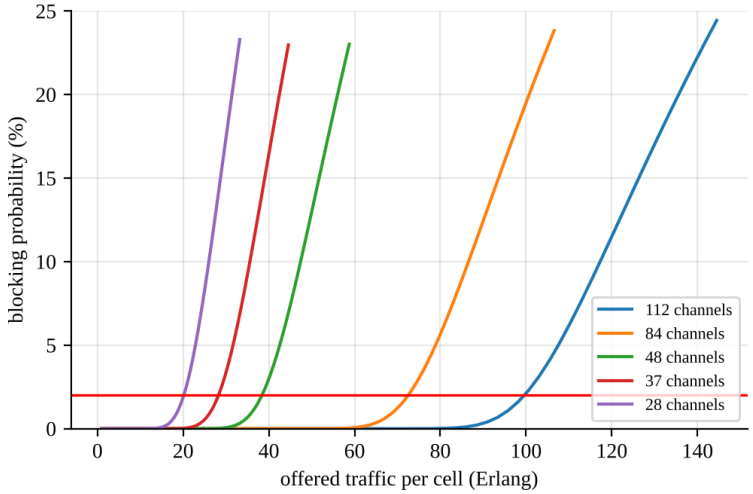


Figure 2: Erlang-B blocking against offered traffic for the channel counts of the strategies; the red line is the 2 % grade of service.

Reading the curves: a 48-channel cell ($N = 7$) carries 38.4 Erlang ($\eta = 80$ %); an 84-channel cell ($N = 4$) carries 72.5 Erlang ($\eta = 86$ %); a 28-channel sector ($N = 4$, 120°) carries only 20.2 Erlang ($\eta = 72$ %), and a 12-channel group carries 6.6 Erlang ($\eta = 55$ %). Efficiency falls as the pool shrinks, which is the trunking cost of sectoring and of large N.

System capacity as a function of N and R. With $M = A_{tot}/(2.598R^2)$ cells,

$$C_{sys}(N, R) = M \cdot A_{cell}(\lfloor S/N \rfloor) = [A_{tot} / (2.598\ R^2)] \cdot B^{-1}(0.02, \lfloor S/N \rfloor)$$

and the largest number of subscribers served is $U_{max} = C_{sys}/a$. When c is large, $A_{cell} \approx c \cdot \eta$ with η slowly increasing, so $C_{sys} \propto 1/(N\ R^2)$. For a sectored layout with s sectors, $C_{sys} = [A_{tot}/(2.598R^2)] \cdot s \cdot B^{-1}(0.02, \lfloor S/(sN) \rfloor)$. The capacity density, C_{sys}/A_{tot}, is what the planner buys with small R and small N.

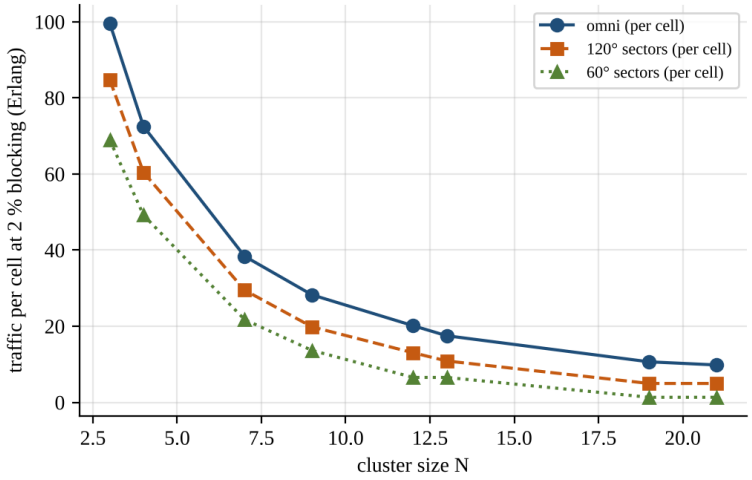


Figure 3: Traffic per cell at 2 % blocking against cluster size for omni, 120° and 60° sectored cells.

3.3 Co-channel SIR for each strategy

S1 (omni). Path loss gives a received power $P_r \propto d^{-n}$. A mobile at the cell edge, at distance R from its own base station, receives the interference of the co-channel cells of the first tier. There are six of them, all approximately D away. Then

$$\mathrm{SIR} = S/I = R^{-n} / \Sigma_{k=1}^{6} D_k^{-n} \approx R^{-n}/(6 \cdot D^{-n}) = (D/R)^n / 6 = (3N)^{n/2} / 6$$

In decibels, $\mathrm{SIR}_{dB} = 10 \cdot \log_{10}[(3N)^{n/2}/6]$. For N = 7, n = 4 this is 441/6 = 73.5, i.e. 18.66 dB. Inverting, the smallest N that meets a threshold γ is

$$N \geq (6\gamma)^{2/n} / 3$$

For γ = 15 dB (31.6) this gives N ≥ 4.59, 6.67, 11.0 and 22.1 for n = 4, 3.5, 3 and 2.5. The next valid cluster sizes are 7, 7, 12 and 25.

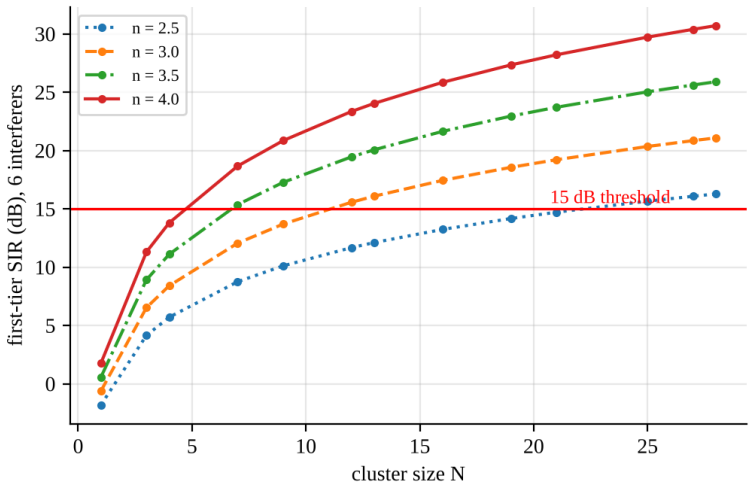


Figure 4: First-tier SIR against cluster size for four path-loss exponents; the 15 dB threshold is the red line.

S2 (sectored). A 120° antenna illuminates a sector of 120°. Of the six first-tier co-channel cells, only those whose antennas point towards the mobile can interfere significantly; by the

geometry of the hexagonal grid this is two cells for 120° sectors and one for 60° sectors. Therefore

$$\mathrm{SIR_s} = Q^n / i_o, \quad i_o = 2\ (120°),\ i_o = 1\ (60°)$$

With a finite front-to-back ratio F (20 dB here) the remaining cells add $(6 - i_o)\cdot 10^{-F/10}$, so the effective count is $i_o + (6 - i_o)/100$, i.e. 2.04 and 1.05. The cluster size that meets 15 dB at n = 3.5 is $N \geq (2\gamma)^{2/n}/3 = 3.56$, i.e. N = 4 for 120° sectors, and $N \geq 2.4$, i.e. N = 3 for 60° sectors. The capacity per cell is $s\cdot B^{-1}(0.02, \lfloor S/(sN) \rfloor)$: 3 × 20.15 = 60.5 Erlang for N = 4 with 120° sectors, against 38.4 Erlang for N = 7 omni, a gain of 57 %.

S3 (splitting). The cell-edge signal after a split is $P'\cdot(R/2)^{-n}$. Requiring it to equal the old cell-edge signal $P\cdot R^{-n}$ gives

$$P'\,(R/2)^{-n} = P\,R^{-n} \implies P' = P / 2^n$$

Since all cells (and therefore all interferers) are reduced in the same ratio, D/R is unchanged and $\mathrm{SIR} = (3N)^{n/2}/6$ exactly as in S1. Capacity rises because the number of cells grows by four in each split cell while the channels per cell stay at S/N. When a small cell is placed inside a large cell of the same plan, the micro-cell mobile sees signal $P'(R/2)^{-n} = P\,R^{-n}$ and the same interference from the surrounding macro cells, so its SIR is also unchanged.

S4 (dynamic). A channel may be reused at any distance $D \geq D_{min}$ that still gives $\mathrm{SIR} \geq \gamma$. From $\mathrm{SIR} = (D/R)^n/6 \geq \gamma$,

$$D_{min} = R\,(6\gamma)^{1/n}, \quad N_{eff} = (D_{min}/R)^2 / 3 = (6\gamma)^{2/n} / 3$$

This is 4.59 at n = 4 (against N = 7 in S1, a capacity potential of 7/4.59 = 1.52), 6.67 at n = 3.5 (1.05) and 11.0 at n = 3 (1.09 against N = 12). S4 therefore gains most where the environment is benign, and also because it avoids the integer rounding of N.

3.4 Planning cost combined with capacity

Let the planning cost be $C_{plan} = c_{sv}\cdot A_{tot} + c_{tr} + c_{pl}\cdot M$ (survey proportional to area, one traffic study, and per-site radio planning), and the capital cost of the sites $M\cdot C_s(R)$ with $C_s(R) = \sigma_s\cdot C_1\cdot\sqrt{R}$, where σ_s = 1 for omni sites and 1.25 for three-sector sites. With $M = A_{tot}/(2.598R^2)$:

$$C_{tot}(R) = c_{sv}\cdot A_{tot} + c_{tr} + [A_{tot}/(2.598\,R^2)]\cdot(c_{pl} + \sigma_s\cdot C_1\cdot\sqrt{R})$$

The operating capacity of the same layout is the system capacity of §3.2, $C_{sys} = [A_{tot}/(2.598R^2)]\cdot s\cdot B^{-1}(0.02, \lfloor S/(sN) \rfloor)$. The two are linked through R, which is the free variable. Eliminating R gives the cost per Erlang

$$\kappa(R) = C_{tot} / C_{sys} = [\ (c_{sv}\,A_{tot} + c_{tr})\cdot 2.598R^2/A_{tot} + c_{pl} + \sigma_s\,C_1\,\sqrt{R}\ ] / [\ s\cdot B^{-1}(0.02, \lfloor S/(sN) \rfloor)\ ]$$

For large systems the fixed term is negligible and $\kappa \approx (c_{pl} + \sigma_s\cdot C_1\cdot\sqrt{R})/A_{cell}$: **the cost per Erlang is the cost per site divided by the traffic per site**. With the stated constants this is $(0.6 + 30\cdot\sqrt{0.26})/38.4 = 0.41$ lakh per Erlang for S1 at 100,000 subscribers (R = 0.26 km). For S2 (N = 4, 120°) it is $(0.6 + 1.25\cdot 30\cdot\sqrt{R})/60.5$ and R is larger because each cell carries more, so it is cheaper per Erlang when demand is uniform. For S3 the extra feature is that the cost of a round is incremental: the cost of splitting is $C(k) = C(k-1) + 3M_{k-1}[c_{pl} + C_1\sqrt{R_k}]$

(Task 6), so the capital is spent only where and when the demand arrives. For S4 the sites are those of the underlying layout and the added cost is the controller software, treated as a fixed amount that does not depend on R, so κ is lowest only when the channel pool is the limiting resource.

Table 5: Summary of the derived functions

Quantity	S1	S2	S3	S4
Cells for 12 km²	$4.62/R^2$	$4.62/R^2$	$5 \cdot 4^k$ (uniform)	as base layout
Channels per cell	$\lfloor S/N \rfloor$	$\lfloor S/(sN) \rfloor$ per sector	$\lfloor S/7 \rfloor = 48$	pooled, up to S
SIR (first tier)	$(3N)^{n/2}/6$	$(3N)^{n/2}/i_0$	unchanged, $(3N)^{n/2}/6$	$\geq \gamma$ by admission test
Traffic per cell	$B^{-1}(c)$	$s \cdot B^{-1}(c)$	$B^{-1}(48)$ per cell	B^{-1} of the pooled set
Cost per Erlang	$(c_{pl} + C_1\sqrt{R})/A_{cell}$	$(c_{pl} + 1.25C_1\sqrt{R})/(s \cdot A_c)$	incremental, Σ over rounds	software + base layout

Answer to Task 4 – Analyse Best, Average and Worst-Case Behaviour (4 Marks)

4.1 SIR: where in the cell is it best, average and worst?

Take the serving base station at the origin and the six first-tier co-channel base stations at distance $D = R\sqrt{(3N)}$. A mobile at distance r from its own base station has signal r^{-n} and interference $\Sigma\, d_k^{-n}$, where d_k is its distance to the k-th interferer.

Best case: close to the base station. As $r \to 0$ the signal grows as r^{-n} while the interference stays at about $6 \cdot D^{-n}$, so $SIR \approx (D/r)^n/6 = (Q \cdot R/r)^n/6$. For $r = 0.1R$, $N = 7$ and $n = 4$ this is $(45.8)^4/6 = 7.3 \times 10^5 = 58.6$ dB. The six interferers are then all far away and almost equidistant, which is why the equidistant approximation is exact in this case; in practice the SIR is limited by receiver noise, not by interference.

Average case: mid-cell. The mean distance from the centre of a regular hexagon to a uniformly distributed point is 0.61R (the median is 0.64R; found by sampling 2 million points), so a typical mobile is at $r \approx 0.61R$. The six interferers are no longer equidistant: the two nearest are about $D - r \cdot \cos\theta$ away and the two farthest about $D + r \cdot \cos\theta$. For $N = 7$ and $n = 4$ the SIR at this radius towards a vertex is 26.3 dB, and the mean of the SIR in dB over all uniformly distributed positions is 28.6 dB (simulation with six interferers, $\sigma = 0$).

Worst case: cell corner. At a vertex, $r = R$. Two interferers are at about $D - R$, two at about D and two at about $D + R$, so

$$SIR_{worst} \approx R^{-n} / [\, 2(D - R)^{-n} + 2D^{-n} + 2(D + R)^{-n} \,]$$

For N = 7, n = 4 (Q = 4.583): $(Q-1)^{-4} = 0.00607$, $Q^{-4} = 0.00227$, $(Q+1)^{-4} = 0.00103$. The denominator is $2 \times (0.00607 + 0.00227 + 0.00103) = 0.01874$, so SIR = 53.4, i.e. **17.3 dB**. The exact geometry of the simulator gives 17.8 dB, and the equidistant approximation gives 18.7 dB. The worst case is therefore about 1 dB below the textbook figure, and the equidistant formula is a good estimate of the worst case only because its optimism about the nearest interferers is partly cancelled by its pessimism about the farthest ones.

Table 6: Deterministic SIR at three positions (three tiers of interferers, no shadowing; N = 4 omni at n = 4 shown for comparison)

N	n	Best r = 0.1R (dB)	Average r = 0.61R (dB)	Worst, corner (dB)	Worst, 120° sectors (dB)
7	4.0	58.0	26.3	17.2	25.1
7	3.5	49.4	21.8	13.9	21.3
7	3.0	40.9	17.2	10.5	17.3
7	2.5	32.2	12.5	6.9	13.3
4	4.0	53.1	21.3	11.8	18.6

Interferer geometry in the three cases. Best: six interferers at equal distance D. Average: two near ($\approx D - 0.7R$), two at D, two far ($\approx D + 0.7R$). Worst: two at $D - R$, which dominate the sum because of the steep power law, since $(D-R)^{-n}/D^{-n} = (Q/(Q-1))^n = 2.7$ for Q = 4.58 and n = 4. With 120° sectors the interferers pointing away from the mobile are attenuated by the front-to-back ratio, and the worst-case SIR rises to 25.1 dB because only the two co-channel sectors that illuminate the corner remain.

4.2 Blocking: at which time of day is it best, average and worst?

Blocking does not depend on position but on the offered traffic per cell compared with the c channels. For an S1 / S3 cell with c = 48 (capacity 38.4 Erlang at 2 %) at the 100,000-subscriber design:

Table 7: Erlang-B blocking of a 48-channel cell under different traffic conditions

Traffic condition	Offered per cell (Erlang)	Blocking B(A, 48)
Off-peak (night, 10 % of busy hour)	3.8	1.9e-33 %
Average hour (50 %)	19.2	1.5e-06 %
Busy hour (100 %, design load)	38.4	2.00 %
Busy hour with +10 % forecast error	42.2	4.70 %
Event surge, mean zone cell (simulation)	94.8	50.37 %
Event surge, hottest zone cell	230.4	79.27 %

The **best** blocking behaviour is off-peak (essentially zero: $B = 10^{-35}$), the **average** hour is also negligible ($B = 1.5 \times 10^{-8}$) because 48 channels have a large reserve at half load, the **design (busy-hour)** point is by definition 2 %, and the **worst** case is the event surge, where the same

cell offered 94.8 Erlang blocks half of the calls and the hottest cell (230 Erlang) blocks about 80 %. The steepness of the Erlang-B curve is the essential observation: a 10 % forecast error (42.2 instead of 38.4 Erlang) multiplies the blocking by about 2.4 (2 % → 4.7 %), so the design needs a margin or an overflow mechanism. For the sectored cell of S2 the same table is shifted: c = 28 and a capacity of 20.2 Erlang per sector, and the curve is slightly less steep.

4.3 Effect of shadowing and path-loss exponent

Path-loss exponent. SIR (dB) = $10 \cdot \log_{10}[(3N)^{n/2}/6]$ is linear in n: each unit of n changes the SIR by $5 \cdot \log_{10}(3N)$ dB, which is 6.6 dB for N = 7. From the table, the worst-case SIR of the N = 7 cell falls from 17.2 dB at n = 4 to 10.5 dB at n = 3 and 6.9 dB at n = 2.5; the 15 dB threshold is not met at the corner for $n \leq 3.5$ (13.9 dB at n = 3.5) and is met there only at n = 4. For the population as a whole the 5th-percentile SIR meets 15 dB for n = 3.5 (15.6 dB) and n = 4 (19.3 dB) but not for n = 3 (12.0 dB) or n = 2.5 (8.2 dB). In the open campus (n between 2.5 and 3) the cluster N = 7 is therefore not adequate, and those cells would need N = 12 (n = 3), N = 25 (n = 2.5) or sectors; this is recorded as a limitation in Task 11.

Log-normal shadowing. If both the desired signal and each interferer carry independent log-normal factors of standard deviation σ (in dB), the SIR in dB is the difference of the desired level and the (approximate) log-sum of the interferers. Its standard deviation is about $\sigma \cdot \sqrt{(1 + 1/k_{eff})}$, where k_{eff} is the number of interferers of comparable strength; for σ = 8 dB and $k_{eff} \approx 6$ this is 8.6 dB. Adding the spread caused by the mobile position (7.7 dB standard deviation in the simulator for N = 7, n = 3.5), the total standard deviation is $\sqrt{(7.7^2 + 8.6^2)} = 11.6$ dB, in agreement with the simulated value of 11.7 dB. At n = 4 the mean SIR falls from 27.9 to 22.6 dB and the 5th percentile falls from 19.3 dB to 4.2 dB.

Outage at the corner. For N = 7, n = 3.5 the corner SIR without shadowing is 13.9 dB, so the margin over 15 dB is -1.1 dB. With σ = 8 dB the outage probability at the corner is $Q((13.9 - 15)/8.6) = Q(-0.13) = 55$ %. To reach a 10 % outage at the corner the median SIR would have to exceed the threshold by 1.28 × 8.6 = 11 dB, i.e. be 12.2 dB higher than it is; since the SIR in dB changes by $17.5 \cdot \log_{10}(N)$ at n = 3.5, raising it by N alone would need N to grow by a factor of 5.0 (to $N \approx 35$); this is why shadowing margin is provided in practice by power control, handoff to the best server and frequency hopping rather than by raising N, and why the simulated outage figures of Task 9 (which have none of these) are an upper bound.

Summary. Best SIR near the base station, average at mid-cell, worst at the corner; best blocking off-peak, worst in the event surge. Shadowing costs 5 dB of mean SIR and 15 dB at the 5th percentile; lowering n from 4 to 2.5 costs 10.3 dB at the corner (N = 7).

Answer to Task 5 – Scaling and Sensitivity Analysis (3 Marks)

5.1 Scaling relations

Table 8: Proportionalities

Quantity	Relation	Strategy where it applies

Quantity	Relation	Strategy where it applies
System capacity	$C_{sys} \propto 1/(N \cdot R^2)$ (c large)	S1; S2 with $s \cdot B^{-1}$ in place of c; S3 with $R = R_0/2^k$
SIR	$SIR \propto N^{n/2} = Q^n$	S1, S3 (N fixed); S2 with $1/i_0$ instead of 1/6
Cells	$M \propto 1/R^2 \propto U$	all; S3 as 4^k
Minimum N for threshold	$N_{min} \propto (i_0\gamma)^{2/n}$	S1, S2; N_{eff} for S4
Transmit power	$P \propto R^n$	S3 ($P_k = P_0/2^{nk}$)
Handoff rate	$H \propto v/R$	all; S3 doubles per round
Cost per Erlang	$\approx (c_{pl} + C_1\sqrt{R})/A_{cell}$	all

Capacity ∝ 1/N comes from channels per cell S/N; **SIR ∝ $N^{n/2}$** comes from $Q^n = (3N)^{n/2}$; together they give the central trade-off, since doubling N multiplies the SIR by $2^{n/2}$ (6.0 dB at n = 4, 5.3 dB at n = 3.5) and halves the capacity of every cell. **Cells ∝ $1/R^2$** and, because demand is proportional to U, $R \propto U^{-1/2}$: a tenfold increase in subscribers needs a $\sqrt{10} = 3.2$ times smaller radius, and a hundredfold increase (1,000 → 100,000) needs a tenfold smaller radius.

Comparison of growth. S1 and S2 add cells continuously in proportion to U (S2 with a constant about 0.64 times that of S1 because each cell carries 60.5 instead of 38.4 Erlang). S3 adds cells in steps of ×4 (uniform) or ×(1 + 3f) (selective), so it overshoots demand by up to a factor of 4 under uniform demand but follows the demand map under clustered demand. S4 does not change the order of growth; it only lowers the constant by at most the factor N/N_{eff}.

5.2 Sensitivity to the path-loss exponent and to growth

Table 9: Minimum cluster size, capacity and number of cells at 100,000 subscribers for different n and sectoring (uniform demand)

n	Layout	N for 15 dB	Channels / sector	Erlang / cell	Cells for 100,000	R (km)
2.5	omni	25	13	7.4	338	0.12
2.5	120°	12	9	13.0	192	0.16
2.5	60°	7	8	21.8	115	0.20
3.0	omni	12	28	20.2	125	0.19
3.0	120°	7	16	29.5	85	0.23
3.0	60°	4	14	49.2	51	0.30
3.5	omni	7	48	38.4	66	0.26
3.5	120°	4	28	60.5	42	0.33
3.5	60°	3	18	68.9	37	0.35
4.0	omni	7	48	38.4	66	0.26

n	Layout	N for 15 dB	Channels / sector	Erlang / cell	Cells for 100,000	R (km)
4.0	120°	3	37	84.8	30	0.39
4.0	60°	3	18	68.9	37	0.35

When n falls from 4 to 2.5 the required omni cluster grows from N = 7 to N = 25, so the cell capacity falls from 38.4 to 7.4 Erlang (81 % less, because 13 channels per cell is a very small pool) and the number of cells needed for 100,000 subscribers rises from 66 to 338, a factor of 5.1. Sectoring helps at every n: at n = 2.5, 120° sectors (N = 12) cut the number of cells from 338 to 192 and 60° sectors (N = 7) cut it to 115; at n = 3 the cells fall from 125 to 85 (120°) and 51 (60°); at n = 4, 120° sectors (N = 3, 37 channels per sector) cut the cells from 66 to 30. At n = 4 the 60° layout (N = 3, only 18 channels per sector) needs 37 cells, more than the 120° layout, because six small channel pools have poor trunking efficiency: beyond three sectors the saving in interference is paid for by a loss of trunking gain.

Growth from 1,000 to 100,000+ subscribers. At 1,000 subscribers (25 Erlang, 2.1 Erlang/km²) the network is coverage-limited: 3 macro cells of R = 1.24 km already have 115 Erlang. At 10,000 the demand (250 Erlang) is 2.2 times the capacity of the coverage layer and radii must fall by about 40 %. At 100,000 (2,500 Erlang, 208 Erlang/km²) the radius is 0.26 km for S1 and the layout needs 66 cells. Beyond 100,000 the same trend gives R = 0.19 km at 200,000 and 0.12 km at 500,000.

5.3 Break-even scales and practical limits

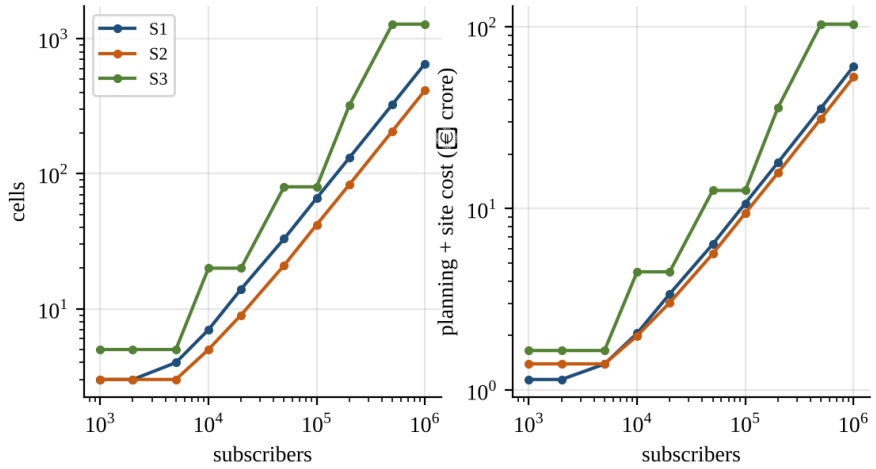


Figure 5: Cells and cost (planning + sites) against subscribers for S1, S2 and S3 under uniform demand (analytic model).

Three break-even points follow from the analysis and the simulation. (1) **Sectored overtakes omni at about 6,200 subscribers** on the uniform-demand cost model: below it the layout is coverage-limited, so the 25 % dearer sector sites buy nothing; above it each sectored site carries 1.57 times the traffic and the total cost is up to 12 % lower (e.g. Rs 9.5 crore against Rs 10.7 crore at 100,000). (2) **The coverage layout stops being enough at about 2,318 subscribers** in the actual township because of the hot spots: the busiest cell of the 1.5 km

layout carries 16.6 Erlang per 1,000 subscribers, which exceeds 38.4 Erlang at about 2,300 subscribers. Beyond this point selective splitting or an equivalent measure is needed, and the simulation shows that at 10,000 subscribers selective splitting (23 sites, Rs 4.6 crore) already costs 23 % less than a uniform layout sized for the hottest cell (27 sites, Rs 6.0 crore) and at 100,000 it costs 45 % less (Rs 19.9 against 36.5 crore). (3) **Splitting stops being practical** when the nominal radius falls below about 60 m, because a base-station site then costs more than the traffic it carries and a vehicle at 30 km/h makes about 9 handoffs per call; in the simulation this occurs only in the last two rounds in the hottest hot spots, which contribute 21 of the 181 sites.

Under **uniform** demand uniform splitting is the worst strategy (its capacity steps are ×4, so the 80 cells at 100,000 subscribers are 21 % more than the 66 cells that S1 needs, and at 200,000 it needs 320 cells against 131). The superiority of S3 therefore depends entirely on the clustered nature of demand, which is why it is evaluated with the township traffic map in Tasks 8 to 10.

Answer to Task 6 – Formulate the Recurrence for Repeated Cell Splitting (3 Marks)

Original problem and subproblem. The original problem (base case, round k = 0) is the coverage layout: $M_0 = \lceil 12/(2.598 \cdot 1^2) \rceil = 5$ cells of radius $R_0 = 1$ km, each with the c = 48 channels of the N = 7 plan, so the capacity of the layout is $C_0 = M_0 \cdot A_{cell} = 5 \times 38.4 = 192$ Erlang. The demand is D = a·U = 25, 250, 500 and 2,500 Erlang for the four forecasts. If $D > C_{k-1}$, each overloaded cell of radius R_{k-1} defines a subproblem: the same problem on a cell of radius $R_k = R_{k-1}/2$, area $A_c/4$ and demand $A_c/4$ times the local density.

Number of new cells per split. A cell of radius R is replaced by four cells of radius R/2 (area 1/4 each). One of them keeps the original base station; **three new base stations are added**. Hence a split turns 1 cell into 4: net +3.

6.1 Uniform splitting

If all cells are split in round k:

$$M_k = 4 \cdot M_{k-1}, \quad M_0 = 5$$

$$C_k = M_k \cdot A_{cell} = 4 \cdot C_{k-1}, \quad C_0 = 5 \times 38.4 = 192 \text{ Erlang}$$

$$R_k = R_{k-1}/2, \quad R_0 = 1 \text{ km}$$

$$P_k = P_{k-1}/2^n, \quad P_0 \text{ given (e.g. 20 W)}$$

$$H_k = 2 \cdot H_{k-1}, \quad H_0 = \text{handoff rate of the macro layout } (\propto v/R)$$

The power recurrence comes from keeping the received edge power constant (Task 3). The handoff recurrence comes from the fact that the time to cross a cell is proportional to R, so the handoff rate per call is proportional to 1/R. Each of the recurrences has the single initial condition at round 0 (the base case: no split performed).

Work (cost) per round. Round k installs $3M_{k-1}$ new sites of radius R_k, so

$$W_k = 3 \cdot M_{k-1} \cdot [c_{pl} + C_1\sqrt{R_k}], \quad T_k = T_{k-1} + W_k, \quad T_0 = c_{sv} \cdot A_{tot} + c_{tr} + M_0(c_{pl} + C_1\sqrt{R_0})$$

With the constants of the cost model, $T_0 = 9.6 + 3 + 5 \times 0.6 + 5 \times 30 =$ Rs 165.6 lakh. The work term has two components: the planning work $3M_{k-1} \cdot c_{pl}$, which grows as 4^k, and the site capital $3M_{k-1} \cdot C_1\sqrt{R_k}$, which grows as $4^k/2^{k/2} = (2\sqrt{2})^k$ and so more slowly, because smaller cells are cheaper.

6.2 Selective splitting

Let n_k be the number of cells that are overloaded at the start of round k, and $f_k = n_k/M_{k-1}$ the overloaded fraction. Only these are split, each creating three new cells:

$$M_k = M_{k-1} + 3 \cdot n_k = M_{k-1}(1 + 3f_k), \quad M_0 = 5 \text{ (or 7 in the rectangular simulation)}$$

$$C_k = C_{k-1}(1 + 3f_k) \quad \text{(capacity where it is needed)}$$

$$P_k(i) = P_0/2^{n \cdot d_i} \quad \text{for a cell that has been split } d_i \text{ times}$$

$$W_k = 3 \cdot n_k \cdot [c_{pl} + C_1\sqrt{R_k}], \quad T_k = T_{k-1} + W_k$$

Uniform splitting is the special case $f_k = 1$ for all k. Because f_k decreases as the hot spots are relieved, the factor $(1 + 3f_k)$ decreases from 4 towards 1 and the splitting stops (base case of the recursion: no overloaded cell or $R < R_{min}$). The recurrence was checked against the simulator, which records the number of overloaded cells in each round (S3, 100,000 subscribers):

Table 10: Recurrence check against the simulator (100,000 subscribers)

Round k	Cells M_k (simulated)	Overloaded n_k	$f_k = n_k/M_{k-1}$	$M_{k-1}(1 + 3f_k)$
0	7	–	–	–
1	19	7	1.000	28
2	55	12	0.632	55
3	118	21	0.382	118
4	160	14	0.119	160
5	178	6	0.037	178
6	181	1	0.006	181

From round 2 to round 6 the recurrence reproduces the simulated number of cells exactly. In round 1 the recurrence predicts 28 cells but the simulation has 19, because the seven macro cells are adjacent and the vertices at which new base stations are placed are shared by neighbouring split cells; a vertex shared by three split cells yields one site, not three. This sharing effect is the only deviation, and it makes the real network cheaper than the recurrence predicts.

Answer to Task 7 – Solve the Recurrence and Interpret the Result (4 Marks)

Method. The recurrences are first-order linear and have constant ratios, so **iteration (repeated substitution)** is the natural method: substituting the recurrence into itself k times produces a pattern that is recognised and then confirmed by induction. The master theorem is not applicable because the recurrences describe growth, not a divide-and-conquer cost T(n) = aT(n/b) + f(n); and a characteristic-equation approach would be unnecessarily heavy for a first-order equation.

7.1 Closed forms for uniform splitting

Iterating $M_k = 4M_{k-1}$: $M_k = 4(4M_{k-2}) = \ldots = 4^k \cdot M_0$. In the same way

$$M_k = M_0 \cdot 4^k, \quad C_k = C_0 \cdot 4^k, \quad R_k = R_0 / 2^k, \quad P_k = P_0 / 2^{nk}, \quad H_k = H_0 \cdot 2^k$$

Proof by induction: true for k = 0; if it holds for k − 1, then $M_k = 4M_{k-1} = 4 \cdot M_0 4^{k-1} = M_0 4^k$. For the cost, iterating $T_k = T_{k-1} + W_k$ with $M_{k-1} = M_0 4^{k-1}$ and $R_i = R_0 / 2^i$ gives

$$T_k = T_0 + \Sigma_{i=1}^{k} 3M_0 4^{i-1}[c_{pl} + C_1 \sqrt{R_0} \cdot 2^{-i/2}] = T_0 + M_0 c_{pl}(4^k - 1) + 3M_0 C_1 \sqrt{R_0} \cdot (\sqrt{2}/2)/(2\sqrt{2} - 1) \cdot [(2\sqrt{2})^k - 1]$$

With $M_0 = 5$, $c_{pl} = 0.6$, $C_1 = 30$ and $R_0 = 1$ this evaluates to

$$T_k = 165.6 + 3.0 \cdot (4^k - 1) + 174.0 \cdot ((2\sqrt{2})^k - 1) \text{ lakh}$$

so T_1 = Rs 492.7 lakh and T_2 = Rs 1,428.6 lakh (Rs 14.3 crore). Building the 80 cells of round 2 directly at R = 0.25 km would cost Rs 12.6 crore, so phasing costs 13 % more in total but only Rs 1.7 crore has to be spent at 1,000 subscribers and Rs 4.9 crore at 10,000; the premium is the price of deferring the investment until demand exists.

7.2 Number of rounds

The capacity after k rounds must carry the demand, $C_k = C_0 4^k \geq D$. Taking logarithms to base 4 and choosing the smallest integer,

$$k = \lceil \log_4 (D / C_0) \rceil \quad \text{(uniform splitting; } k = 0 \text{ if } D \leq C_0\text{)}$$

Selective counterpart. If a fixed fraction f of the cells is overloaded in every round, $C_k = C_0(1 + 3f)^k$, so

$$k = \lceil \log_{1+3f} (D / C_0) \rceil = \lceil \ln(D/C_0) / \ln(1 + 3f) \rceil$$

The two formulas coincide when f = 1 (1 + 3f = 4). In general f decreases from round to round, so the sum of $\ln(1 + 3f_i)$ must reach $\ln(D/C_0)$; the simulated S3 needs six rounds at 100,000 subscribers.

Table 11: Uniform-splitting rounds needed (C_0 = 192 Erlang, P_0 = 20 W, v = 30 km/h, 120 s call)

Subscribers	D (Erlang)	D/C_0	k	Cells	R_k (m)	P_k (W), n = 3.5	Handoffs per call

Subscribers	D (Erlang)	D/C₀	k	Cells	R_k (m)	P_k (W), n = 3.5	Handoffs per call
1,000	25	0.13	0	5	1,000	20.00	0.6
10,000	250	1.30	1	20	500	1.77	1.2
20,000	500	2.60	1	20	500	1.77	1.2
100,000	2,500	13.02	2	80	250	0.16	2.3
200,000	5,000	26.05	3	320	125	0.01	4.6
500,000	12,500	65.12	4	1280	62	0.00	9.2

The table shows that the forecasts of 1,000, 10,000, 20,000 and 100,000 subscribers need 0, 1, 1 and 2 rounds (5, 20, 20 and 80 cells; R = 1, 0.5, 0.5 and 0.25 km). At 200,000 the third round is required, and at 500,000 the demand ratio is 65.1, just above $4^3 = 64$, so a fourth round (1,280 cells, R = 62.5 m) would be forced by a margin of 1.7 %; this shows how coarse the quantisation by a factor 4 is.

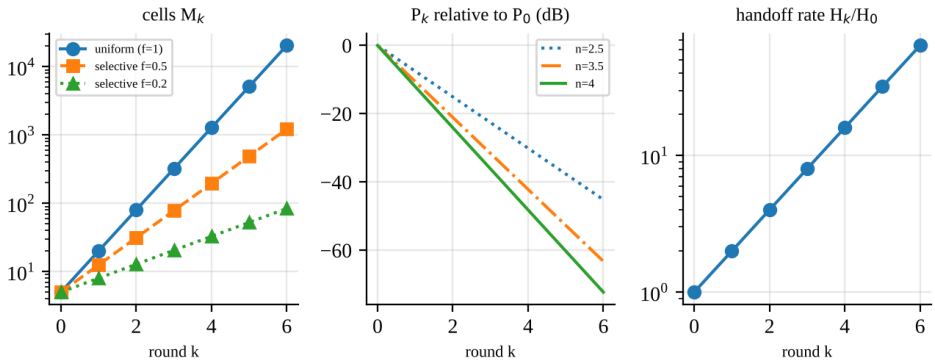


Figure 6: Growth of cells (uniform and selective), reduction of transmit power and increase of handoff rate over the rounds.

7.3 Interpretation

- **Growth in sites.** The number of cells grows as $4^k = (D/C_0)\cdot(1 \text{ to } 4)$: linear in demand, with an overshoot of up to a factor of 4 for uniform splitting, which is the cost of the coarse quantisation. The cost grows more slowly, as $(2\sqrt{2})^k = 2.83^k$, because the cells get cheaper as they shrink, and in the selective case only the f-fraction is paid for.
- **Reduction of transmit power.** $P_k = P_0/2^{nk}$: −12 dB per round at n = 4 and −10.5 dB at n = 3.5. A 20 W macro transmitter falls to 1.77 W after one round and 0.16 W after two, which lowers the cost of amplifiers and the interference footprint, and allows handset batteries to last longer.
- **Handoff rate.** $H_k = 2^k \cdot H_0$. A vehicle moving at 30 km/h makes about 0.6 handoffs per 120 s call in a 1 km cell, 2.3 in a 250 m cell and 9.2 in a 62.5 m cell. Handoffs consume signalling capacity and cause dropped calls, so there is a limit to splitting: the cells along the arterial road must remain larger (or use an umbrella cell) while the stationary hostel population can be served by the smallest cells.

- **Conclusion.** The closed form says that capacity can be bought with one design variable, R, but at the price of linearly growing sites, falling power and doubling mobility load per round; the recurrence is therefore valid only while the handoff load remains manageable, which restricts the practical number of rounds to about four ($R \geq 62.5$ m).

Answer to Task 8 – Implement and Test the Planning Strategies (4 Marks)

8.1 Software structure

The strategies were implemented in Python 3 with NumPy and SciPy in a single module, cellplan.py (the key functions are listed in the Annexure). Each building block of the analysis has its own function so that every number in this report can be traced to code.

Table 12: Components of the simulator

Component	Function	What it does
Erlang-B calculator	erlang_b(A, c), cap(c)	Exact recursion B(A,k); bisection for the traffic that c channels carry at 2 % blocking
Cluster geometry	valid_cluster_sizes, ij_of, cochannel_positions	Valid $N = i^2 + ij + j^2$; coordinates of the first-tier (or 18 nearest) co-channel cells using the (i, j) move
Hex-grid generator	hex_lattice(R)	Base-station lattice of spacing $\sqrt{3}\cdot R$ covering the 4 km × 3 km township
Channel assignment	units_of, evaluate	Nearest-site (and sector) assignment of each subscriber; channels per unit = $\lfloor S/(N\cdot s) \rfloor$; offered traffic and Erlang-B blocking per unit
SIR evaluator (Monte-Carlo)	sir_mc, sir_at	SIR of mobiles dropped uniformly in a hexagon, with path loss n, optional log-normal shadowing and sector pattern (20 dB front-to-back)
Subscriber generator	make_subscribers	Clustered township demand (Section 8.2), nested datasets from one seed
Planners	find_cluster_size, design_uniform, design_selective	S1 (iterative search), S2 (sector loop) and S3 (recursive cell splitting)
Dynamic assignment	call_sim	Event-driven call-level simulation with fixed assignment or borrowing from neighbouring cells (S4)

8.2 Datasets

Subscriber positions are drawn once for 100,000 subscribers with a fixed seed (2026) and the smaller datasets are the first 1,000, 10,000 and 20,000 points of the same draw, so that all strategies see exactly the same subscribers at every scale and the results are directly comparable. The distribution is a mixture that reflects the township description:

Table 13: Synthetic township demand model

Component	Share	Shape (km)
Uniform background (open campus, roads, other)	35 %	uniform over 4 × 3
Academic blocks	20 %	Gaussian, centre (1.2, 1.5), $\sigma = 0.35$
Hostels	20 %	Gaussian, centre (2.8, 1.0), $\sigma = 0.30$
Residential colonies	15 %	Gaussian, centre (3.2, 2.3), $\sigma = 0.40$
Arterial road	10 %	uniform in x along y = 1.5, $\sigma_y = 0.05$

The demand is therefore strongly non-uniform (in the 85-cell average-sized S1 layout the busiest cell is offered 145 Erlang against an average of 29), which is what makes selective splitting interesting. Every subscriber offers 0.025 Erlang. The same subscribers are used in the 'average-sized' designs (R fixed by total capacity) and in the 'peak-sized' designs (R reduced until no cell is overloaded).

8.3 Test cases

The test matrix required by the task was run in full: N = 3, 4, 7, 12 with n = 2.5, 3, 4, each with omni (1 sector, 6 interferers), 120° (3 sectors, 2 interferers) and 60° (6 sectors, 1 interferer) layouts. Each case uses 100,000 mobiles dropped uniformly in the reference cell and the three tiers of 18 nearest co-channel cells. The analytic column is $(3N)^{n/2}/i_0$ in dB (the equidistant approximation).

Table 14: SIR test matrix (σ = 0, three tiers of interferers)

N	n	Layout	Analytic (dB)	Simulated mean (dB)	5th percentile (dB)	Outage < 15 dB (%)
3	2.5	omni	4.1	8.8	3.3	88.0
3	2.5	120°	8.9	14.5	9.7	67.1
3	2.5	60°	11.9	17.3	12.7	40.8
3	3.0	omni	6.5	12.7	6.0	73.3
3	3.0	120°	11.3	18.9	13.1	29.3
3	3.0	60°	14.3	21.8	16.3	0.0

N	n	Layout	Analytic (dB)	Simulated mean (dB)	5th percentile (dB)	Outage < 15 dB (%)
3	4.0	omni	11.3	20.2	11.1	34.6
3	4.0	120°	16.1	27.4	19.7	0.0
3	4.0	60°	19.1	30.5	23.1	0.0
4	2.5	omni	5.7	10.4	5.0	83.9
4	2.5	120°	10.5	14.7	9.8	65.7
4	2.5	60°	13.5	16.9	12.3	45.7
4	3.0	omni	8.4	14.6	8.1	64.5
4	3.0	120°	13.2	19.2	13.3	26.8
4	3.0	60°	16.2	21.5	16.0	0.1
4	4.0	omni	13.8	22.8	14.0	13.0
4	4.0	120°	18.6	27.9	20.0	0.0
4	4.0	60°	21.6	30.4	23.1	0.0
7	2.5	omni	8.7	13.5	8.2	71.6
7	2.5	120°	13.5	19.1	14.2	16.8
7	2.5	60°	16.5	22.2	17.4	0.0
7	3.0	omni	12.1	18.4	12.0	37.8
7	3.0	120°	16.8	24.3	18.4	0.0
7	3.0	60°	19.8	27.4	21.7	0.0
7	4.0	omni	18.7	27.9	19.3	0.0
7	4.0	120°	23.4	34.3	26.5	0.0
7	4.0	60°	26.4	37.5	30.0	0.0
12	2.5	omni	11.7	16.5	11.2	51.2
12	2.5	120°	16.4	21.5	16.5	0.0
12	2.5	60°	19.5	24.2	19.3	0.0
12	3.0	omni	15.6	22.0	15.6	1.6
12	3.0	120°	20.3	27.2	21.3	0.0
12	3.0	60°	23.3	30.0	24.2	0.0
12	4.0	omni	23.3	32.8	24.2	0.0
12	4.0	120°	28.1	38.5	30.6	0.0
12	4.0	60°	31.1	41.5	33.7	0.0

8.4 Correctness checks

- **Known case.** For N = 7 and n = 4 the equidistant approximation gives $(21)^2 / 6 = 73.5 = 18.66$ dB; the code reproduces 18.66 dB. The exact-geometry evaluation at the cell corner

with six interferers gives 17.82 dB, which differs by 0.84 dB, as predicted analytically in Task 4.

- **Geometry.** The six nearest co-channel cells of the N = 7 pattern computed by the code are at distances 4.583, 4.583, 4.583, 4.583, 4.583, 4.583 (in units of R), equal to $\sqrt{21}$ = 4.583 as required by $D = R\sqrt{(3N)}$.
- **Hex lattice.** A lattice of radius 0.5 km has 27 sites and one of radius 1.5 km has 7 sites; these include the partially covered border cells, so they exceed the $4.62/R^2$ area estimate (18.5 and 2.1) by the border effect.
- **Reproducibility.** Two runs with the same seed give identical arrays (checked: True); all seeds are fixed in the code.
- **Erlang-B.** The closed form was checked against an independent event-driven simulation of a single 48-channel cell (validation table below).
- **Channel conservation.** For every N, $N \cdot \lfloor S/N \rfloor \leq 336$ (e.g. $7 \times 48 = 336$, $4 \times 84 = 336$, $12 \times 28 = 336$, $19 \times 17 = 323$).

Table 15: Validation of the Erlang-B formula for c = 48 channels

Offered traffic (Erlang)	Erlang-B (%)	Simulated (%)	Calls simulated
19.2	0.0000	0.0000	25,620
30.0	0.0602	0.0300	40,035
38.4	1.9991	1.9589	51,254
48.0	10.6788	10.2193	64,173
60.0	24.3599	24.0606	80,372

The simulated blocking agrees closely with the formula at every load (within 0.5 percentage points; for example 1.96 % against 2.00 % at the design point), so the call-level model and the analytic model are consistent and the Erlang-B assumption of Poisson arrivals with exponential holding is reproduced by the simulator. The cell-edge, N = 7 and n = 4 check is the 'known case' requested in the task.

Answer to Task 9 – Record and Present Experimental Performance (3 Marks)

The measurements are the SIR distribution (mean and 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time. Strategies S1, S2, S3 and the hybrid S2+S3 were designed for each of the four subscriber scales; S4 is evaluated in the call-level simulator. 'Average-sized' means the lattice radius is chosen so that the total capacity of all cells equals the total demand; 'peak-sized' means the radius is reduced until no cell exceeds its Erlang-B capacity.

9.1 Layout and blocking

Table 16: Layout and blocking, average-sized (S1, S2) and selectively split (S3, S2+S3) designs

Strategy	Subscribers	Sites	Mean R (km)	Channels per unit	Blocking (%)	Overloaded units	Design time (s)
S1	1,000	7	1.500	48	0.00	0/7	0.00
S1	10,000	7	1.500	48	47.20	1/7	0.00
S1	20,000	19	0.600	48	20.00	5/19	0.02
S1	100,000	85	0.270	48	25.40	20/85	0.05
S2	1,000	7	1.500	28	0.00	0/21	0.00
S2	10,000	7	1.500	28	35.07	4/21	0.00
S2	20,000	11	0.950	28	32.33	9/33	0.01
S2	100,000	45	0.375	28	29.72	42/135	0.05
S3	1,000	7	1.000	48	0.00	0/7	0.00
S3	10,000	23	0.446	48	0.16	0/23	0.01
S3	20,000	50	0.273	48	0.00	0/50	0.02
S3	100,000	181	0.127	48	0.03	0/181	0.17
S2+S3	1,000	7	1.000	28	0.00	0/21	0.00
S2+S3	10,000	18	0.528	28	0.20	0/54	0.01
S2+S3	20,000	50	0.285	28	0.16	0/150	0.02
S2+S3	100,000	187	0.127	28	0.06	0/561	0.22

Reading the table. At 1,000 subscribers every strategy gives zero blocking with the 7-site coverage layout. At 10,000 the average-sized S1 and S2 layouts have only 7 sites and block 47 % and 35 % of calls, because the busiest cell is offered 166 Erlang (S1) and the busiest sector 66 Erlang (S2), against capacities of 38.4 and 20.2 Erlang: the average-sized designs are blind to the hot spots. At 100,000 the average-sized S1 layout (85 sites) still blocks 25 % because its busiest cell carries 145 Erlang. The selectively split designs reach 0.03 % (S3) and 0.06 % (S2+S3), because splitting continues until no cell is overloaded. 'Mean R' for S3 and S2+S3 is the average over a mixture of radii, from 1 km in the quiet areas down to 1/64 km = 16 m at the centre of the hot spots.

9.2 Peak-sized uniform designs

Table 17: Uniform lattices sized for the hottest cell (blocking below 2 % everywhere)

Strategy	Subscribers	Sites	R (km)	Blocking (%)	Planning (Rs cr)	Sites (Rs cr)	Total (Rs cr)	Time (s)
S1	1,000	7	1.500	0.000	0.17	2.57	2.74	0.00
S1	10,000	27	0.505	0.312	0.29	5.76	6.04	0.30
S1	20,000	59	0.310	0.171	0.48	9.85	10.33	0.68

Strategy	Subscribers	Sites	R (km)	Blocking (%)	Planning (Rs cr)	Sites (Rs cr)	Total (Rs cr)	Time (s)
S1	100,000	331	0.120	0.046	2.11	34.40	36.51	4.21
S2	1,000	7	1.500	0.000	0.17	3.21	3.38	0.00
S2	10,000	23	0.575	0.172	0.26	6.54	6.80	0.35
S2	20,000	45	0.375	0.148	0.40	10.33	10.73	0.86
S2	100,000	233	0.150	0.051	1.52	33.84	35.36	5.40

This is the fair comparison with selective splitting: the uniform layout must be as fine as the hottest hot spot demands. At 100,000 subscribers S1 needs 331 sites (R = 120 m) and S2 needs 233 sites (R = 150 m). Sectoring therefore saves 30 % of sites against the omni design but, since a three-sector site is 25 % dearer, only 3 % of the cost (Rs 35.4 crore against Rs 36.5 crore).

9.3 Cost of the four strategies

Table 18: Planning plus site cost (Rs crore) of GoS-satisfying designs

Subscribers	S1 peak-sized	S2 peak-sized	S3 selective	S2+S3 selective
1,000	2.74	3.38	2.27	2.79
10,000	6.04	6.80	4.64	5.03
20,000	10.33	10.73	7.83	10.06
100,000	36.51	35.36	19.91	25.28

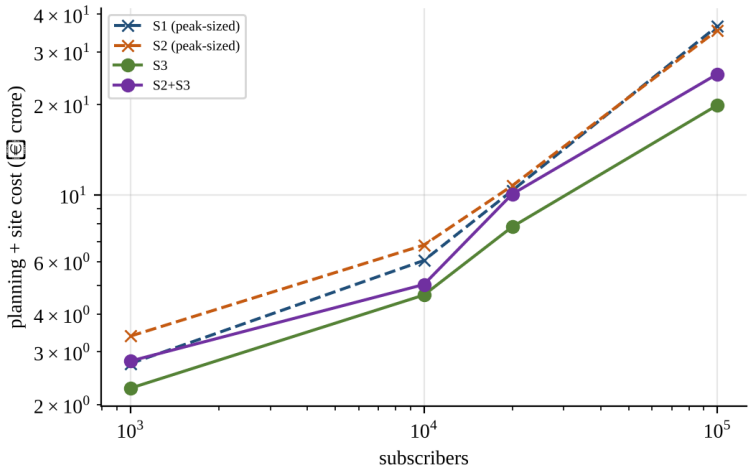


Figure 7: Planning plus site cost against subscribers for the GoS-satisfying designs of the previous table.

9.4 SIR measurements

Table 19: SIR statistics of the two channel plans (200,000 samples, 15 dB threshold)

Configuration (n = 3.5)	Analytic (dB)	Mean σ = 0	5 % σ = 0	Outage σ = 0 (%)	Mean σ = 8	5 % σ = 8	Outage σ = 8 (%)
S1/S3: N = 7, omni	15.4	23.2	15.6	2.0	17.7	0.1	42.8
S2/S2+S3: N = 4, 120°	15.9	23.6	16.7	0.0	18.8	1.2	38.3

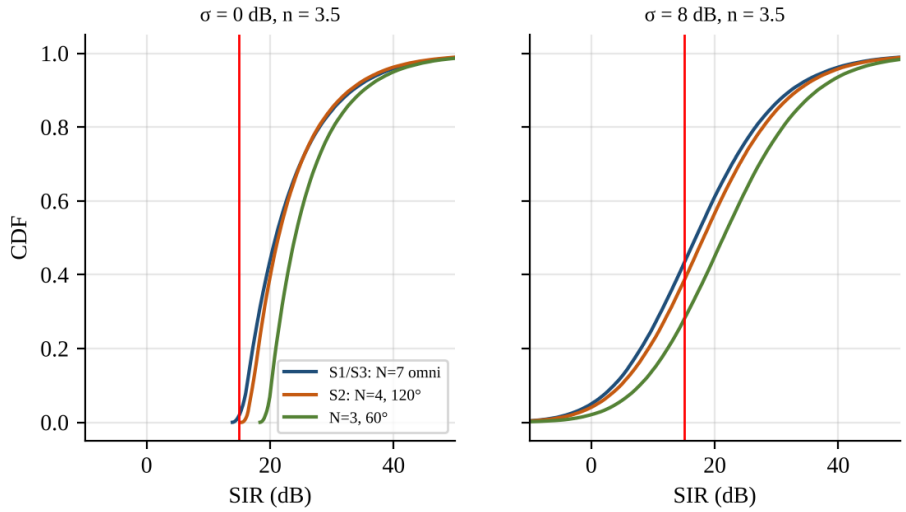


Figure 8: SIR cumulative distribution functions without shadowing (left) and with σ = 8 dB (right).

Without shadowing the omni N = 7 plan has an outage of 1.9 % against 15 dB and the sectored N = 4 plan has none, with 5th percentiles of 15.6 dB and 16.7 dB, just above the threshold. With σ = 8 dB, both plans fail the threshold in 38–43 % of positions, which is the reason the margin discussion of Tasks 4 and 11 matters.

9.5 Blocking against scale

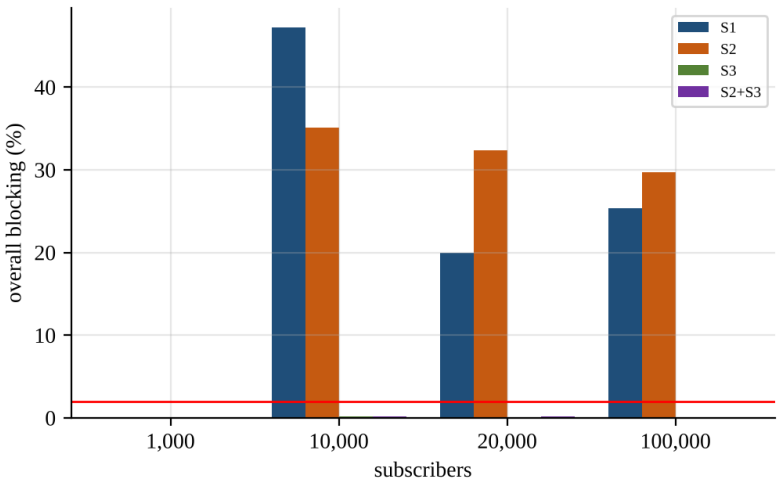


Figure 9: Overall blocking of the four average-sized or split designs; the red line is the 2 % target.

9.6 Capacity against N and call-level behaviour of S4

The capacity against N figure (Figure 4) and the Erlang-B curves (Figure 3) in Task 3 are the other two presentations requested. For S4, a 19-cell hexagonal patch (N = 7 plan, 48 channels per cell, 15 % of each neighbour's channels reserved for the owner) was simulated for 60 mean holding times with Poisson arrivals. Core cells are the seven central cells; the event load is the stadium surge (1,458 Erlang spread over the core, 31.2 Erlang elsewhere).

Table 20: Blocking with fixed assignment and with channel borrowing (S4), 19-cell patch

Load	Core, fixed (%)	Core, borrowing (%)	All cells, fixed (%)	All cells, borrowing (%)
Normal busy hour (31.2 E per cell)	0.08	0.00	0.12	0.00
Half event load	60.38	44.56	41.66	31.09
Full event load	79.71	71.89	65.00	58.88

Borrowing cuts normal-load blocking from 0.12 % to zero and trims the surge blocking, but when the neighbours are themselves close to full, as under the surge, the gain is limited (core blocking 79.7 % → 71.9 % at the full event load, 60.4 % → 44.6 % at half load). Dynamic assignment is therefore a good complement to a good layout and not a substitute for capacity, which supports its role in Task 11 as a remedy for the last overloaded cells.

9.7 Run-time

Designing and evaluating S1 or S2 at 100,000 subscribers takes under 0.3 s for the average-sized lattice, while the iterative peak-sizing search (one evaluation per candidate radius,

0.005 km steps) takes 4.2 s for S1 and 5.4 s for S2. Selective splitting with six rounds takes only 0.17 s for S3 and 0.22 s for S2+S3 because each round re-evaluates only the cells it creates (vectorised nearest-site search); the SIR Monte-Carlo with 100,000 samples takes 0.02 s. Planning effort in the simulator is thus negligible compared with the surveying and site work that the cost model accounts for.

Answer to Task 10 – Compare Theory with Simulation (4 Marks)

10.1 SIR

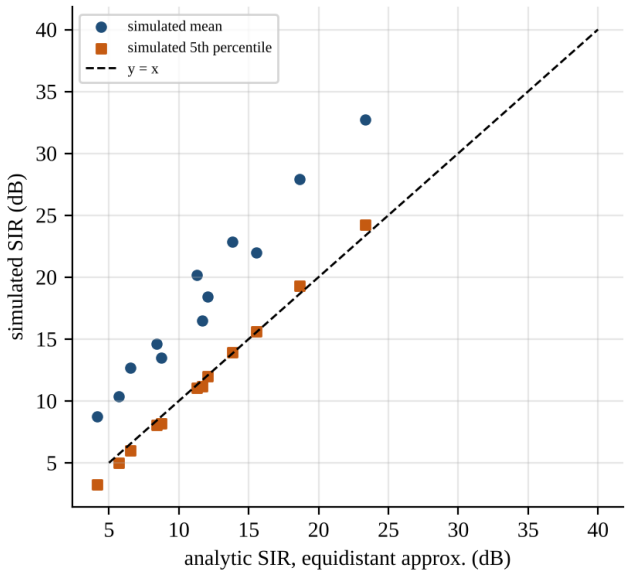


Figure 10: Simulated mean and 5th-percentile SIR against the analytic equidistant-interferer value for all omni test cases (N = 3, 4, 7, 12; n = 2.5, 3, 4).

The analytic value $(3N)^{n/2}/6$ sits **below the simulated mean in every case, by 5–10 dB, and within about ±1 dB of the simulated 5th percentile** (for example N = 7, n = 4: analytic 18.7 dB, simulated 5th percentile 19.3 dB, mean 27.9 dB; N = 7, n = 2.5: analytic 8.7 dB, 5th percentile 8.2 dB). The formula is therefore not an estimate of the average SIR but a good estimate of the worst case, which is exactly what it is derived from (a mobile at the cell edge). In the Task 4 geometry the corner SIR is 17.2 dB at N = 7, n = 4, which is 1.5 dB below the formula and 2 dB below the simulated 5th percentile, because the 5 % of mobiles nearest the corner are weighted by their small probability.

Sectoring. The ideal analytic gain for 120° sectors is $10 \cdot \log_{10}(6/2) = 4.8$ dB at fixed N. The simulated gain at N = 4, n = 4 is 27.9 − 22.8 = 5.1 dB in the mean and 20.0 − 14.0 = 6.0 dB at the 5th percentile, which is a bit more than predicted because the mobiles near the sector edges also profit from the 20 dB front-to-back ratio.

10.2 Sources of deviation (with sizes)

Table 21: Why the simulation differs from the analysis

Source	Effect found in this study

Source	Effect found in this study
Equidistant interferers	Theory is 1–1.5 dB above the true corner SIR and 5–10 dB below the mean; equal-distance assumption puts the mobile at the corner and all interferers at D
Finite tiers	Using six interferers instead of 18 raises the mean by 0.7 dB (N = 7, n = 4) and up to 1.5 dB at n = 2.5; second-tier cells matter more when n is low
Log-normal shadowing	σ = 8 dB lowers the mean SIR by 5.3 dB and the 5th percentile by 15 dB (N = 7, n = 4: 27.9 → 22.6; 19.3 → 4.2), and raises the outage from 0 to 27 %
Mobile-position distribution	Uniform positions in the hexagon give a mean 9 dB above the corner formula; clustered users at the base (hostels) do better, users at the edge worse
Poisson arrivals, infinite sources	Erlang-B matches the call-level simulation within 0.5 percentage points of blocking (validation table in Task 8); with about 1,500 subscribers per cell the finite-source (Engset) correction is small, so this is the smallest deviation
Integer N and channels	336/19 = 17.7 → 17 channels per cell loses 13 channels (3.9 %), and N = 7 vs N_{eff} = 6.67 wastes 5 % of capacity in S1 (what S4 recovers)
Hexagon covering a rectangle	The 4 × 3 km township needs 85 cells at R = 0.27 km in the simulator against 66 from $12/(2.598R^2)$, +29 % because of partially covered border cells
Non-uniform demand	Uniform-demand analysis predicts 2 % blocking at 100,000 subscribers for 66 cells; the simulation gives 25 % blocking at 85 cells because the busiest cell carries 4.9 times the average cell load (145 against 29 Erlang)

10.3 Blocking and sites

Table 22: Number of sites: analytic against simulated

Strategy	Subscribers	Sites, uniform theory	Sites, average-sized sim.	Sites, peak-sized sim.
S1	10,000	7	7	27
S1	100,000	66	85	331
S2	10,000	5	7	23
S2	100,000	42	45	233
S3	10,000	20	23	–
S3	100,000	80	181	–

The analytic uniform-demand count (66 for S1) is a **lower bound** that the real township exceeds by a factor of 5 for omni cells (331 sites) when the layout is uniform, and by about 2.7 for S3 (181 sites against the analytic 66 of S1 at equal average demand). Theory and simulation therefore agree on the trend (sites ∝ subscribers) and on the SIR formula for the worst case, but disagree by a large factor on the number of sites, and the disagreement is entirely caused by the non-uniform demand map, which the analytic model does not include.

10.4 Which strategy scales best?

Judged by the number of sites and the cost at 100,000 subscribers, **S3 (selective cell splitting)** scales best: 181 sites and Rs 19.9 crore, against 331 sites and Rs 36.5 crore for S1, 233 sites and Rs 35.4 crore for S2, and 187 sites and Rs 25.3 crore for S2+S3. Both pieces of evidence support this: the analysis says splitting keeps SIR constant while adding capacity only where needed (Task 3, 6), and the simulation shows that this reduces the sites by 45 % against the uniform design. S2+S3 has the highest SIR margin (5th percentile 16.7 against 15.6 dB without shadowing) but costs 27 % more, so its extra margin has to be paid for. S4 does not scale on its own, but improves every other strategy at its overloaded cells (Section 9.6). On uniform demand the order is different (S2 best, uniform splitting worst, Task 5), so the verdict depends on the demand map, which is why the demand map must be measured before the strategy is chosen.

Answer to Task 11 – Recommend and Defend the Final Design (4 Marks)

11.1 Recommended design

Recommendation: S3 with S4 as overflow. Deploy a hierarchical macro–micro network that starts from the 7-site macro layer (R_0 = 1 km) with the N = 7 channel plan (6 carriers = 48 channels per cell, 38.4 Erlang per cell at 2 % blocking), omnidirectional antennas, and grows by **selective** cell splitting only where the measured blocking exceeds 2 %. Splitting is limited to four rounds ($R \geq 62.5$ m, Task 7). The few cells that remain overloaded are relieved by borrowing from neighbouring cells (S4) and, if necessary, by 120° sectors on those few sites.

Table 23: Recommended growth path with simulated results (n = 3.5, a = 0.025 E)

Subscribers	Rounds	Sites	Smallest R	Blocking (%)	Overloaded cells	Cost (Rs crore)
1,000	0	7	1 km	0.00	0	2.27
10,000	3	23	125 m	0.16	0	4.64
20,000	4	50	62.5 m	0.00	0	7.83
100,000 (4 rounds, practical cap)	4	160	62.5 m	1.17	6 of 160	18.25
100,000 (6 rounds, unrestricted)	6	181	15.6 m	0.03	0 of 181	19.91

With the four-round limit the 100,000-subscriber network has 160 sites, an overall blocking of 1.17 % (within the 2 % target on average), 6 overloaded cells out of 160 (3.8 %) and costs Rs 18.25 crore, Rs 1.66 crore less than the unrestricted design, which would add 21 further sites with radius below 62.5 m (down to 16 m at the very centre of the hostels). The SIR is that of the N = 7 plan: 15.4 dB at the cell edge (n = 3.5), simulated mean 23.2 dB, 5th percentile 15.6 dB, outage 1.9 % without shadowing. Transmit power falls by $2^{3.5} = 11.3$ times per round, so, starting from a 20 W macro cell, the transmitter is about 0.16 W after two rounds and a few milliwatts after four.

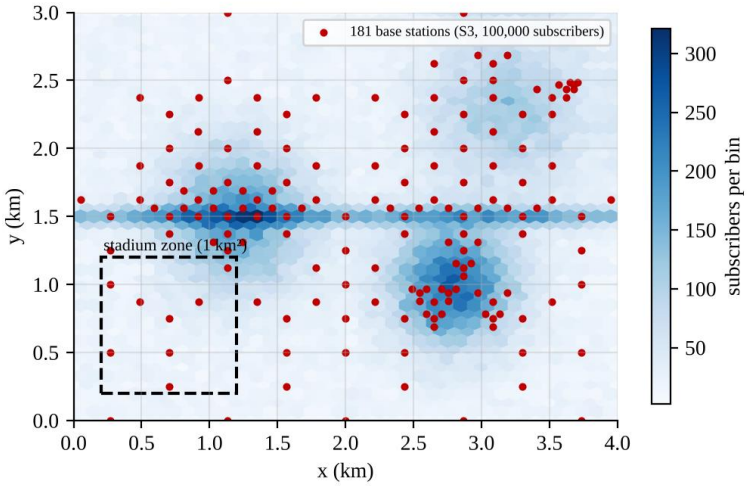


Figure 11: Demand density and the 181 base stations of the unrestricted S3 design at 100,000 subscribers (the 160-site recommended design omits the three finest rounds in the hot spots).

11.2 Why it beats the alternatives

Table 24: Comparison of the designs at 100,000 subscribers

Design (100,000 subscribers)	Sites	Cost (Rs cr)	Blocking (%)	5th pct SIR (dB) / outage	Remark
S1 uniform, peak-sized	331	36.51	0.05	15.6 / 1.9 %	Simple, but 1.8× the sites of S3; wastes capacity in the quiet 80 % of the township
S2 uniform, peak-sized	233	35.36	0.05	16.7 / 0 %	Fewer sites but 25 % dearer sites; 3 antennas per site; trunking efficiency 72 % against 80 %
S2+S3 selective	187	25.28	0.06	16.7 / 0 %	Best SIR margin; 27 % dearer than S3; three times the control channels
S3 selective (6 rounds)	181	19.91	0.03	15.6 / 1.9 %	Cheapest; but 21 cells below 62.5 m

Design (100,000 subscribers)	Sites	Cost (Rs cr)	Blocking (%)	5th pct SIR (dB) / outage	Remark
Recommended S3, 4 rounds + S4	160	18.25	1.17	15.6 / 1.9 %	Cheapest practical; growth in steps; overflow by borrowing

The choice is not simply the highest capacity: the uniform S1 and S2 designs are able to reach the same blocking but need 1.5 to 2 times as many sites because they must size every cell for the hottest spot. S2+S3 has the best SIR but its additional margin costs Rs 5.4 crore, and as Section 9.4 showed it does not cure the shadowing outage either (38 % against 43 %), since neither design includes power control or handoff to the best server. S4 alone cannot create capacity (Section 9.6). The recommended design is the cheapest design that meets the 2 % target on average, that adds capacity in the right place, and that postpones investment (Rs 2.3 crore at 1,000 subscribers, Rs 4.6 crore at 10,000, Rs 7.8 crore at 20,000).

11.3 Trade-offs and limitations

- **Trunking efficiency against interference.** N = 7 at 80 % trunking efficiency is the smallest omni plan that meets 15 dB at n = 3.5; N = 4 with 120° sectors has only 72 % efficiency per sector but gains 57 % per site (60.5 against 38.4 Erlang). The recommended omni plan keeps the better trunking efficiency, and sectors are kept in reserve for the hottest sites.
- **Handoff load against capacity.** Every round doubles the handoff rate (Task 7); cells along the arterial road should therefore be kept large or overlaid with an umbrella cell, which the simulator does not model (the road carries only 10 % of the subscribers).
- **Shadowing.** With σ = 8 dB and no power control the outage against 15 dB is about 43 %. The deployed system relies on power control, handoff to the best server and a 6–10 dB fade margin in the link budget, none of which was simulated; the SIR statistics in this report are therefore pessimistic for a practical system but are valid for comparison of the strategies.
- **Open-campus cells.** The N = 7 plan is sized for n = 3.5. In the open campus (n between 2.5 and 3) its 5th-percentile SIR is only 8.2–12.0 dB, so cells there need N = 12 or sectors if the campus is to be served with the same quality; this refinement was not simulated.
- **Model limits.** The demand map is synthetic, the cost model uses stated unit prices (a different price per site changes the absolute numbers, although the ranking was not re-tested for other prices), no mobility or traffic-channel reassignment is simulated, and Erlang-B assumes blocked calls cleared.

11.4 Beyond 100,000 subscribers

Sites scale roughly in proportion to demand (this is an extrapolation, since only up to 100,000 subscribers were simulated), so doubling the subscribers would need of the order of twice as many sites. The limit is close because at 100,000 the practical four-round cap already leaves 6 overloaded cells. Beyond about 100,000 subscribers the hybrid S2+S3 becomes the logical

next step (the three-sector split triples the capacity of the hottest 62.5 m cells without a smaller radius), then 60° sectoring (68.9 Erlang per cell at n = 2.5 against 38.4), and finally new spectrum, since spectrum cannot be increased in this assignment. The design therefore stays practical in its present form to about 100,000 subscribers, and beyond that needs the hybrid and finally additional spectrum.

Answer to Task 12 – Re-evaluate and Adapt the Design (8 Marks)

Decision: MODIFY. The original design (S3, N = 7, selective splitting) cannot carry the event, but it does not need to be redone: the surge sits in one square kilometre. A dedicated sectored micro-cell layer (N = 4, 120° sectors, R = 125 m, 33 sites) over the stadium zone removes the overload (0.85 % blocking) at a lower annual cost than the splitting redesign (Rs 1.20 crore a year if permanent, against Rs 1.71 crore), and the redesign was also rejected because it needs cells as small as 8 m. The parts below give the evidence.

A. Impact Analysis (2 Marks)

New offered traffic density

The attendees offer 25,000 × 0.05 = 1,250 Erlang into 1 km². The residents who normally use the zone offer 122.7 Erlang (simulation, same 100,000-subscriber dataset), so the event-day density in the zone is 1,372.7 E/km². Against the design values:

Table 25: Event density against design density

Reference density	Erlang per km²	Event density ÷ reference
Township average at 100,000 subscribers (2,500 E / 12 km²)	208.3	6.6
Same zone on a normal day (simulated)	122.7	11.2
Attendee traffic alone	1,250.0	6.0
Capacity of the 16 cells serving the zone (16 × 38.4 E)	614.3	2.23

The event density is 6.6 times the township average and 11.2 times the same zone on a normal day, and it is 2.2 times the whole capacity of the zone (1,373 E offered against 614 E of capacity). Each attendee also offers twice the average subscriber (0.05 against 0.025 Erlang), which is the reason a crowd is worse than the same number of ordinary subscribers.

Assumptions that no longer hold

- **Uniform traffic.** The model of Task 3 divides demand evenly; the surge is a point source. Uniform-demand formulas (for example the 66-cell estimate) are meaningless for the event, and only the simulation with the real demand map is valid.
- **Steady demand.** Erlang-B at 2 % is a busy-hour steady-state result. A four-hour peak lasting twice a month (96 hours a year, 1.1 % of the year) is a transient that the design basis never anticipated.

- **Permanent capacity.** Every site in the plan is built for demand that is present all year. Capacity that is only needed for 1.1 % of the year has a very low utilisation if it is built permanently, which changes the economics (Part C).
- **Fixed demand map.** The subscriber distribution was fitted to residents; a 25,000-person crowd arriving in the zone is not part of it. The crowd is also mobile at the beginning and the end of the event, which adds a handoff and location-update peak.

Why the short duration and predictability matter. Because the event is predictable (date, place, size) a planned, temporary response is possible, and because it is brief the response must be judged by its annual cost, not by its capital cost; if the surge were permanent a permanent redesign would be the only option, and if it were unpredictable only a dynamic mechanism (borrowing) would work.

B. Evaluation of the Original Design (2 Marks)

The original design (S3, 181 sites, N = 7) was re-run with the 25,000 attendees added to the same 100,000 subscribers (Erlang-B on every cell, 48 channels per cell, capacity 38.4 Erlang).

Table 26: Behaviour of the original design on event day

Quantity (event day)	Result
Zone offered traffic	1,372.7 Erlang
Cells serving the zone	16
Mean offered load per zone cell / hottest cell	94.8 / 230.4 Erlang (capacity 38.4)
Overloaded cells in the whole network	12 of 181
Blocking in the zone, retained S3	60.7 %
Blocking in the whole township, retained S3	23.1 %
Blocking in the zone on a normal day	0.011 %
Blocking in the zone, if S1 (uniform, 85 sites) were retained	72.3 %
Blocking in the zone, if S2+S3 were retained	45.7 %

What fails first. The traffic channels of the 16 zone cells: the average zone cell is offered 95 Erlang against a capacity of 38.4 (2.5 times), and the busiest 230 Erlang (6 times). Roughly 61 out of every 100 calls attempted in the zone are blocked, and, because the blocked fraction retries, the control channels are overloaded as well. The rest of the township is hardly affected: the 12 overloaded cells are all in or next to the zone, and the whole-township blocking of 23 % is almost entirely the zone's. A uniform S1 layout would be worse (72 %), because it has the larger cells that the crowd falls into.

Interference. The SIR of a cell depends on the geometry (D/R and n) and not on the number of calls; the SIR calculations of Tasks 3, 4 and 9 assume that every co-channel cell is fully loaded, which is exactly the situation of the zone cells during the event. The SIR therefore does not degrade further, and the event is a blocking problem, not a co-channel problem.

What does worsen is the signalling load (paging, location updates, handoffs of the arriving and departing crowd).

Can the design be reused unchanged? No: a blocking probability of 61 % is thirty times the target. **Does an alternative strategy handle the surge more economically?** Three were tested. (i) S2+S3 (sectored) as the base design reduces zone blocking only to 46 %. (ii) Dynamic borrowing (S4, call-level simulation of a 19-cell patch loaded like the event) cuts the core blocking from 79.7 % to 71.9 %, because neighbours of an overloaded cell are overloaded too (at half the event load, from 60.4 % to 44.6 %). (iii) Adding capacity locally by splitting or micro-cells (Part C) is the only one that reaches 2 %.

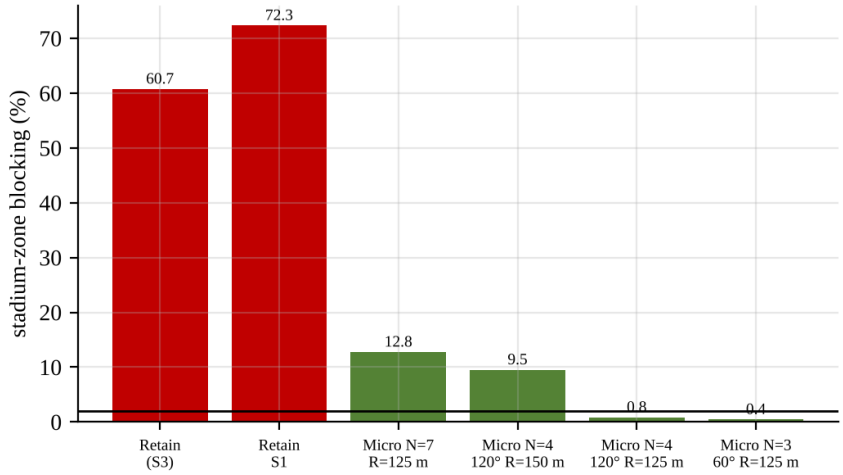


Figure 12: Blocking in the stadium zone on event day for the retained designs and for each micro-cell option (black line: 2 % target).

C. Adaptation or Redesign (2 Marks)

Four ways of adding capacity in the zone were simulated with the same attendees (Erlang-B per sector, the micro layer placed on a hexagonal lattice centred on the zone, SIR constraint 15 dB at n = 3.5).

Table 27: Micro-cell options for the stadium zone (1,373 Erlang offered)

Micro-cell layer	Sites	Capacity (Erlang)	Blocking (%)	Overloaded units	Capital (Rs cr)
N = 7, omni, R = 125 m	33	1,267	12.76	20	3.70
N = 7, omni, R = 62.5 m	114	4,377	0.000	0	9.23
N = 4, 120°, R = 150 m	27	1,632	9.52	44	4.08
N = 4, 120°, R = 125 m	33	1,995	0.85	7	4.57
N = 4, 120°, R = 100 m	45	2,720	0.005	0	5.61
N = 3, 60°, R = 125 m	33	2,275	0.44	2	–

The omni N = 7 layer needs R = 62.5 m (114 sites) before blocking disappears, because 48-channel omni cells carry only 38.4 Erlang each; at R = 125 m (33 sites) it blocks 12.8 %. Sectoring is what makes a sparse layer possible: the 120° N = 4 layer with the same 33 sites

carries 1,995 Erlang (45 % more than offered) and blocks 0.85 %. At R = 150 m (27 sites) the capacity of 1,632 Erlang only just exceeds the offered 1,373 Erlang and, because the load per sector is uneven (up to 31 Erlang against 20.2), 9.5 % of the calls are blocked: this shows that total capacity above demand is not enough. The 60° layer gives the lowest blocking (0.44 %) but its cost is not estimated because the cost model covers only omni and three-sector sites.

Choice among the options

Table 28: Options compared (annual cost = capital × (capital recovery 0.163 + 10 % operation) for permanent options)

Option	Sites added (gross)	Zone blocking (%)	Capital (Rs cr)	Annual cost (Rs cr)	Comment
RETAIN S3	0	60.7	0	0	Fails the 2 % target thirtyfold
MODIFY: temporary 120° layer, 33 sites (cells-on-wheels)	33	0.85	4.57	1.65	Rental 1.5 % of site capital per event day (assumption), 24 events
MODIFY: permanent 120° micro layer, 33 sites	33	0.85	4.57	1.20	Replaces the zone macro cells; idle 98.9 % of the time
REDESIGN: S3 re-split for event load	84	0.08	6.51	1.71	Cells down to 8 m radius

Cost model for the comparison. Permanent equipment is annualised over 10 years at 10 % (capital-recovery factor 0.1627) plus 10 % of capital per year for site rent, power and backhaul. Temporary cells-on-wheels are assumed to cost 1.5 % of the capital value of a site per event day (transport, power, backhaul and a crew), a figure I assumed because none is given; its effect is shown in the sensitivity analysis below. The capital of a micro-cell site is $C_1\sqrt{R}\cdot 1.25 + 0.6$ lakh, so 33 sites at R = 125 m cost Rs 4.57 crore and 45 sites at R = 100 m cost Rs 5.61 crore.

Table 29: Temporary against permanent micro layer (33 sites, 24 events a year)

Rental per event day (% of site capital)	Annual cost, temporary (Rs cr)	Annual cost, permanent (Rs cr)	Break-even events per year
1.0 %	1.10	1.20	26.3
1.5 %	1.65	1.20	17.5
2.0 %	2.20	1.20	13.1
3.0 %	3.29	1.20	8.8

The two modifications are of the same order of cost: at the assumed 1.5 % rental the permanent layer costs Rs 1.20 crore a year against Rs 1.65 crore for temporary cells at 24 events a year, and the break-even is 17.5 events a year. Temporary deployment is therefore cheaper only if events are rarer than about 18 a year or rental is cheaper than about 1.1 % of capital per event day. With the events of this scenario (24 a year) the permanent layer is about 27 % cheaper per year, but the saving disappears if rental is below about 1.1 % of capital per event day, so the decision also rests on the other factors: the temporary layer needs logistics, spectrum re-planning and engineering crews twice a month and permits changes of venue; the permanent layer is simpler to operate, also relieves the zone on normal days, but locks the capital into one place.

The **redesign** (S3 re-split on the event load) is rejected because it costs more per year than the permanent micro layer (Rs 1.71 crore against Rs 1.20 crore) and on practical grounds as well: it adds 84 sites, down to R = 8 m, i.e. cells smaller than a classroom, with a handoff every few seconds for a walking user, in order to serve 1.1 % of the year. Its cost estimate of Rs 6.51 crore is also uncertain at those radii since the cost model floors the radius at 60 m.

Final decision

MODIFY. Install a 120° sectored micro layer of 33 sites (N = 4, R = 125 m, 28 channels per sector) in the stadium zone. Start with temporary cells-on-wheels for the first events (they also show the real attendance, the real path-loss exponent and the real crowd distribution); convert to a permanent layer if events keep their frequency of about two a month, because the break-even above is then crossed. Keep dynamic borrowing (S4) between the micro cells to absorb the 7 sectors that are 8 % over capacity (21.9 Erlang against 20.2), and add the 45-site R = 100 m layer if the attendance forecast exceeds 25,000 by more than about 10 %, since it has no overloaded sector (0.005 % blocking).

Effect on reuse pattern, power, SIR and handoffs

- **Reuse pattern.** The zone is served by the N = 4, 120° plan (28 channels per sector) instead of N = 7 omni (48 per cell); the zone's own channels are reassigned to the micro layer and the macro cells inside the zone are switched off or re-tuned, so the surrounding N = 7 pattern is not disturbed.
- **SIR.** For n = 3.5 the N = 4, 120° plan has an analytic first-tier SIR of 15.9 dB and a simulated mean of 23.6 dB (5th percentile 16.7 dB, outage 0 % without shadowing). **Risk:** in an open stadium (n close to 3) the same plan has a 5th percentile of only 13.3 dB and an outage of 27 %, so a field measurement of n at the first event decides whether the 60° layout (outage 0.1 % at n = 3) is needed; it needs about 20–30 % more sites because its capacity per site is 49 Erlang.
- **Power.** Relative to a macro cell of the retained design, halving the radius cuts the required transmit power by $2^{3.5} = 11.3$ ($P' = P/2^n$, Task 3), so the 125 m micro cells need about one eleventh of the power of cells of twice that radius, a few watts at most, and their footprint is limited to the stadium.
- **Handoffs.** At R = 125 m a walking attendee (5 km/h) crosses a cell in about 1.7 minutes, so handoff rate is moderate; the main handoff load is at the arrival and departure of the crowd, which is predictable, and can be pre-planned by neighbour lists.

Effect on capacity, SIR and planning cost (summary). Capacity of the zone rises from 614 to 1,995 Erlang (×3.2, 45 % above the offered 1,373 Erlang); SIR stays at the design value of 15.9 dB (analytic, n = 3.5); the extra planning is a survey of the zone (the township survey is already paid), a re-plan of the zone's channels and the 33 sites at Rs 4.57 crore (permanent) or about Rs 6.9 lakh per event day (temporary).

D. Evidence-Based Defence (2 Marks)

The decision follows from the evidence in the preceding parts, in four steps.

Table 30: Logic of the decision

Step	Evidence	Consequence
1. Retaining fails	Zone blocking 60.7 % (S3), 45.7 % (S2+S3), 72.3 % (S1); loads 2.5–6 times capacity	RETAIN is rejected
2. Dynamic assignment alone is not enough	Borrowing cuts core blocking only from 80 % to 72 % (neighbours are full)	Capacity must be added
3. Sectored micro cells give the cheapest sufficient capacity	120° N = 4 layer: 33 sites, 0.85 % blocking; omni needs 114 sites; R = 150 m fails (9.5 %)	Sectoring and R = 125 m chosen
4. Redesign costs more than it is worth	+84 permanent sites, radii down to 8 m, used 1.1 % of the year; a permanent micro layer costs Rs 1.20 crore a year against Rs 1.65 crore for temporary cells at 24 events a year (assumed rental)	MODIFY, temporary first, permanent if events persist

The strength of this decision is that it uses the quantities that the earlier tasks produced: the Erlang-B capacity of a cell, the SIR plan N = 4 with 120° sectors derived in Task 3 and tested in Task 8, the demand map of Task 8, the cost model of Task 3, and the finding of Task 9 that adding capacity where the demand is costs far less than adding it everywhere. The weakness is that the cost comparison depends on assumed unit prices and on the real path-loss exponent in the stadium; the sensitivity table and the n = 3 risk statement above show how the decision would change.

Conclusion

A hexagonal cellular network for the 12 km² township was designed, analysed and simulated. The cluster-size, Erlang-B and SIR relations were derived; the best, average and worst cases were located; the recurrence of cell splitting was formulated, solved and checked against a simulator. For the clustered demand of a university township selective cell splitting with N = 7 beats the uniform alternatives by 45 % in sites and cost, with sectoring and dynamic borrowing as complements. The Day-4 surge is met by a sectored micro-cell layer over the stadium zone, temporary at first, permanent if events persist. The main limits are the

synthetic demand map, the assumed prices, and the lack of power control in the SIR simulation.

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ANNEXURE – Source Code (key functions of cellplan.py)

The listing gives the key functions of the planning and simulation module used for Tasks 8 to 12: the Erlang-B calculator, the co-channel geometry, the Monte-Carlo SIR evaluator, the planners for S1 to S3 and the call-level simulator for S4. The numerical results in this report were generated with this code using fixed random seeds.

```
S, W, H, A_SUB, GOS = 336, 4.0, 3.0, 0.025, 0.02   # channels, township km,
Erlang/subscriber, 2 % GoS

def erlang_b(A, c):
    B = 1.0
    for k in range(1, c + 1):
        B = A * B / (k + A * B)
    return B

def cap(c, gos=GOS):
    """Offered traffic (Erlang) a group of c channels carries at blocking <= gos."""
    if c <= 0:
        return 0.0
    key = (c, gos)
    if key not in _cap:
        lo, hi = 0.0, 2.0 * c + 10
        for _ in range(60):
            mid = (lo + hi) / 2
            if erlang_b(mid, c) > gos:
                hi = mid
            else:
                lo = mid
        _cap[key] = lo
    return _cap[key]

def cochannel_positions(N, R, K=18):
    i, j = ij_of(N)
    a1 = np.array([math.sqrt(3) * R, 0.0])
    a2 = np.array([math.sqrt(3) * R / 2, 1.5 * R])
    u = i * a1 + j * a2
    th = math.pi / 3
    rot = np.array([[math.cos(th), -math.sin(th)], [math.sin(th), math.cos(th)]])
    v = rot @ u
    pts = np.array([m * u + k * v for m in range(-4, 5) for k in range(-4, 5) if (m, k) != (0,
0)])
    d = np.hypot(pts[:, 0], pts[:, 1])
    idx = np.argsort(d, kind="stable")
```

```
    return pts[idx[:K]]

def sir_mc(N, n, sectors=1, sigma=0.0, samples=100000, seed=1, K=6, R=1.0,
fb_db=20.0, rmin=0.02):
    rng = np.random.default_rng(seed)
    m = sample_hex(R, samples, rng)
    d0 = np.maximum(np.hypot(m[:, 0], m[:, 1]), rmin * R)
    co = cochannel_positions(N, R, K)
    sig = d0 ** (-n)
    if sigma > 0:
        sig = sig * 10 ** (rng.normal(0, sigma, samples) / 10)
    I = np.zeros(samples)
    ang = np.arctan2(m[:, 1], m[:, 0]) % (2 * np.pi)
    beta = 2 * np.pi / sectors
    centre = (np.floor(ang / beta) + 0.5) * beta
    for p in co:
        v = m - p
        d = np.maximum(np.hypot(v[:, 0], v[:, 1]), 1e-9)
        g = np.ones(samples)
        if sectors > 1:
            a = np.arctan2(v[:, 1], v[:, 0])
            diff = np.abs((a - centre + np.pi) % (2 * np.pi) - np.pi)
            g = np.where(diff <= beta / 2, 1.0, 10 ** (-fb_db / 10))
        pw = d ** (-n) * g
        if sigma > 0:
            pw = pw * 10 ** (rng.normal(0, sigma, samples) / 10)
        I += pw
    return 10 * np.log10(sig / I)

def find_cluster_size(n, gamma_db, i0=6, limit=60):
    """Iterative strategy A: test valid N = i^2+ij+j^2 in increasing order."""
    g = 10 ** (gamma_db / 10)
    for N in valid_cluster_sizes(limit):
        if (3 * N) ** (n / 2.0) / i0 >= g:
            return N
    return None

def hex_lattice(R, reach=0.5):
    dx, dy = math.sqrt(3) * R, 1.5 * R
    pts = []
    rows = int(H / dy) + 4
    for r in range(-rows, rows + 1):
        y = H / 2 + r * dy
        off = (r % 2) * dx / 2
```

```
        for k in range(-rows * 3, rows * 3 + 1):
            x = W / 2 + k * dx + off
            ex = max(0 - x, 0, x - W)
            ey = max(0 - y, 0, y - H)
            if math.hypot(ex, ey) <= reach * R:
                pts.append((x, y))
    return np.array(pts)

def evaluate(BS, subs, sectors, ch, weights=None):
    idx, unit = units_of(BS, subs, sectors)
    nunits = len(BS) * sectors
    w = np.full(len(subs), A_SUB) if weights is None else weights
    A = np.bincount(unit, weights=w, minlength=nunits)
    B = np.array([erlang_b(a, ch) for a in A])
    off = A.sum()
    c = cap(ch)
    return dict(A=A, B=B, blocking=float((A * B).sum() / off), offered=float(off),
                over=int((A > c).sum()), nunits=nunits, idx=idx, unit=unit)

def design_uniform(subs, N, sectors):
    """Strategies A and B: shrink R of a regular lattice until total capacity >= demand."""
    ch = S // (N * sectors)
    need = len(subs) * A_SUB
    per = sectors * cap(ch)
    R = 1.5
    while R > 0.05:
        BS = hex_lattice(R)
        if len(BS) * per >= need:
            return BS, R, ch
        R = round(R - 0.005, 4)
    return hex_lattice(R), R, ch

def design_selective(subs, N, sectors, R0=1.0, max_rounds=8):
    """Strategies C and E: start from radius R0 and split only overloaded cells."""
    ch = S // (N * sectors)
    BS = hex_lattice(R0)
    rad = np.full(len(BS), R0)
    trace = [(0, len(BS), 0)]
    for k in range(1, max_rounds + 1):
        ev = evaluate(BS, subs, sectors, ch)
        c = cap(ch)
        over_units = np.where(ev["A"] > c)[0]
        if len(over_units) == 0:
            break
```

```
        over_bs = np.unique(over_units // sectors)
        new = {}
        for i in over_bs:
            for a in (30, 150, 270):
                p = BS[i] + rad[i] * np.array([math.cos(math.radians(a)),
math.sin(math.radians(a))])
                new[(round(p[0], 3), round(p[1], 3))] = rad[i] / 2
        rad = rad.copy()
        rad[over_bs] /= 2
        known = {(round(p[0], 3), round(p[1], 3)) for p in BS}
        add = [(k_, r_) for k_, r_ in new.items() if k_ not in known]
        if add:
            BS = np.vstack([BS, np.array([a for a, _ in add])])
            rad = np.concatenate([rad, np.array([r for _, r in add])])
        trace.append((k, len(BS), len(over_bs)))
    return BS, rad, ch, trace

def call_sim(loads, c, nbr, borrow, T=120.0, seed=7, reserve=0.15):
    """Poisson arrivals, exponential holding (mean 1). Returns per-cell blocking."""
    rng = np.random.default_rng(seed)
    arr = []
    for i, a in enumerate(loads):
        n = rng.poisson(a * T)
        t = np.sort(rng.uniform(0, T, n))
        arr.append(np.c_[t, np.full(n, i)])
    arr = np.vstack(arr)
    arr = arr[np.argsort(arr[:, 0])]
    hold = rng.exponential(1.0, len(arr))
    busy = np.zeros(len(loads), int)
    dep = []
    tried = np.zeros(len(loads))
    blocked = np.zeros(len(loads))
    warm = 0.1 * T
    for (t, ci), h in zip(arr, hold):
        ci = int(ci)
        while dep and dep[0][0] <= t:
            _, lender = heapq.heappop(dep)
            busy[lender] -= 1
        count = t >= warm
        if count:
            tried[ci] += 1
        if busy[ci] < c:
            busy[ci] += 1
            heapq.heappush(dep, (t + h, ci))
```

```
        continue
    ok = False
    if borrow:
        best, bj = c, -1
        for j in nbr[ci]:
            if busy[j] < best and busy[j] <= c * (1 - reserve) - 1:
                best, bj = busy[j], j
        if bj >= 0:
            busy[bj] += 1
            heapq.heappush(dep, (t + h, bj))
            ok = True
    if not ok and count:
        blocked[ci] += 1
return blocked / np.maximum(tried, 1), tried
```